ScienceUtsav®
Science is awesome

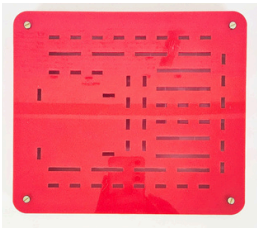
I'm a ROBO SCIENTIST

# SPEED, BRIGHTNESS AND MOTION CONTROL

Session

5

# STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1


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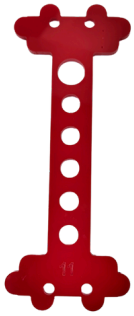


1

Spacers

2

L Clamp

3

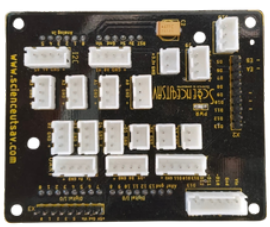
Spacer Screws

4

Nuts

5

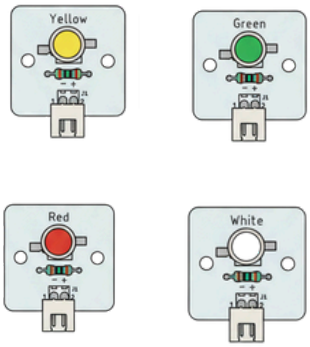
Screw

6

Arduino With Shield

7

Arduino cable

# ELECTRONIC COMPONENTS




**1 Watt LED PCB**

1 Watt LED is a high-brightness light-emitting diode that produces much stronger light compared to normal small LEDs.

**DIP LED**

A DIP LED (Dual In-line Package LED) is a type of LED with two long metal pins used for easy connection on breadboards and PCB circuits.

**DC MOTOR**

A DC motor is an electromechanical device that converts direct current (DC) electrical energy into rotational mechanical motion. It is commonly used in automation systems, robotics, fans, pumps, and various motor-driven applications to provide controlled movement and power.

**Potentiometer PCB**

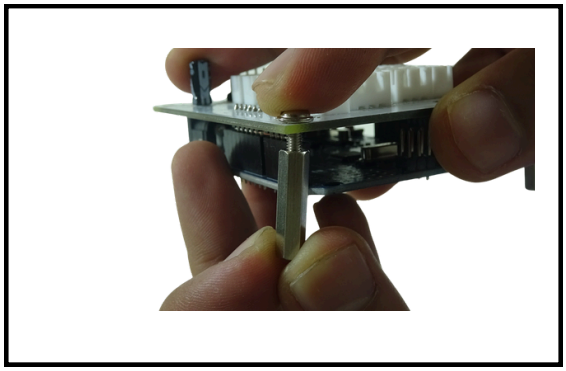
A potentiometer is a variable resistor that allows the adjustment of electrical resistance within a circuit. It is commonly used for controlling voltage levels, adjusting signal strength, and providing user input in electronic devices such as audio systems, control panels, and instrumentation equipment.

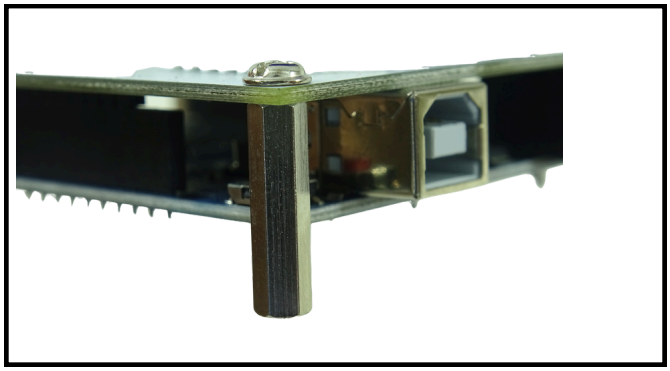
**2 PIN RMC**

Used to connect the 1-watt LEDs and the buzzer.

# Procedure :

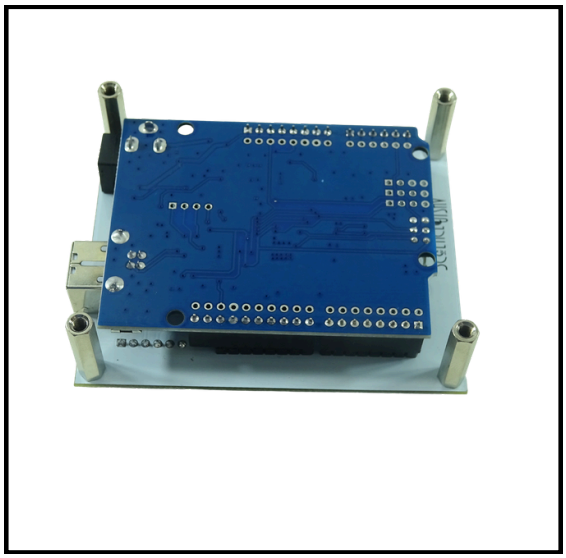
**Step 1 :** Fit the spacer to Arduino Shield as shown below using Spacer screw

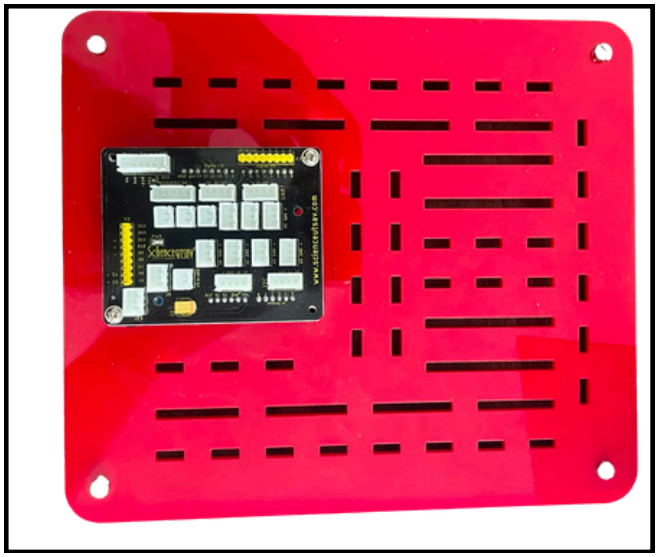




**Step 2 :** Repeat the step 1 for all 4 corners of Arduino Shield

**Step 3 :** Place the Arduino Shield to the base board as shown below

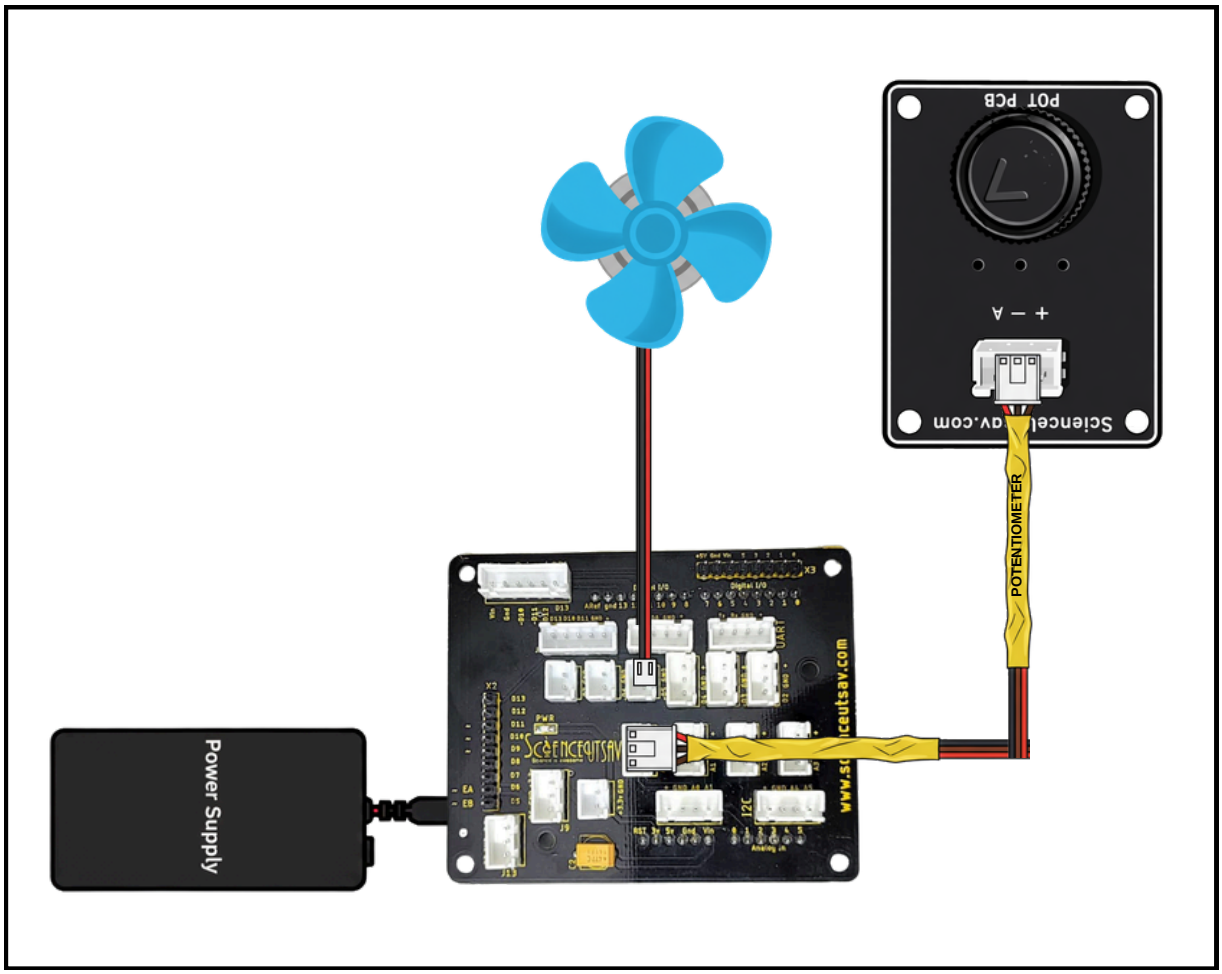




**Step 4 :** Fix it with the spacer screw from the bottom as shown below

# BEGINNER PROJECT

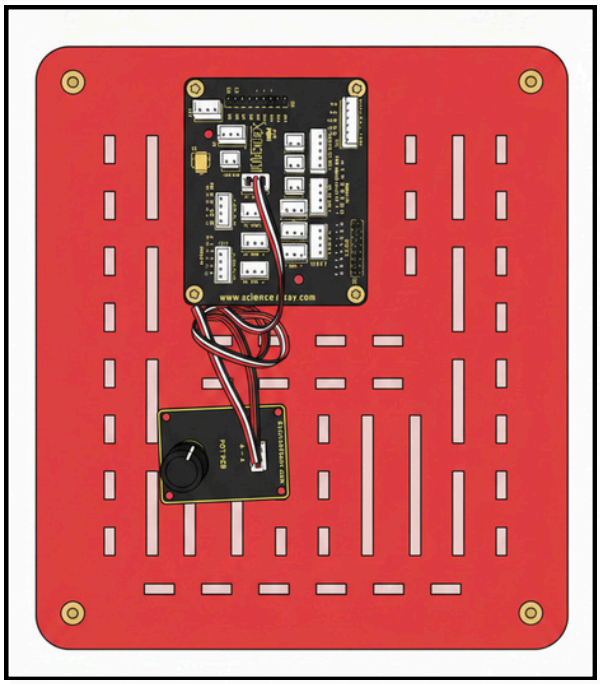
The circuit diagram of the project is given below



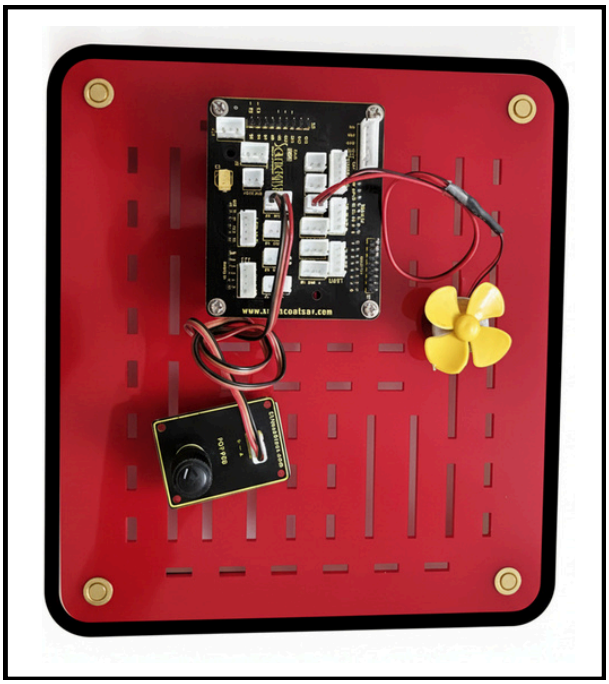

## CONNECTION:

1. DC MOTOR IN D5
2. POTENTIOMETER PCB IN A0

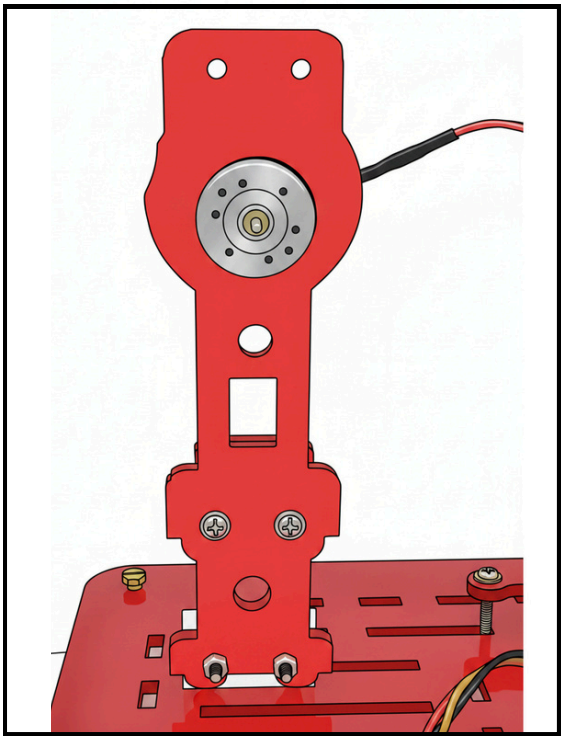
**Step 1-** Fix the potentiometer PCB to the board



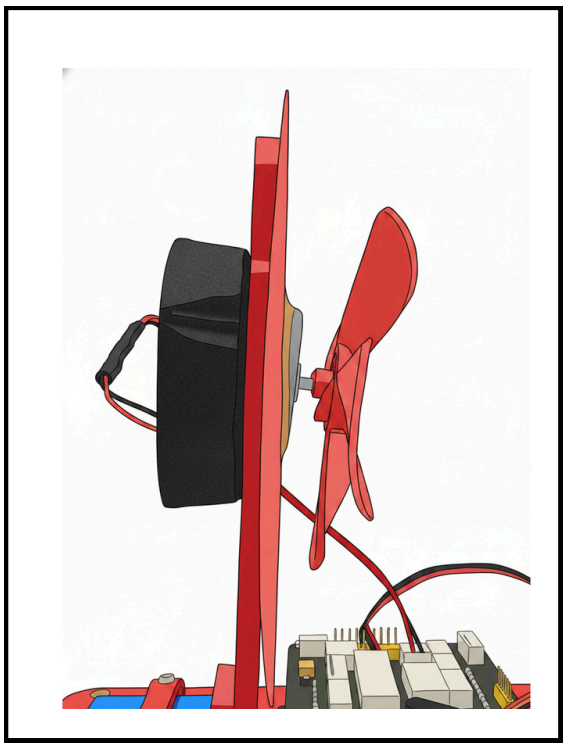
**Step 2-** Connect the dc motor as shown



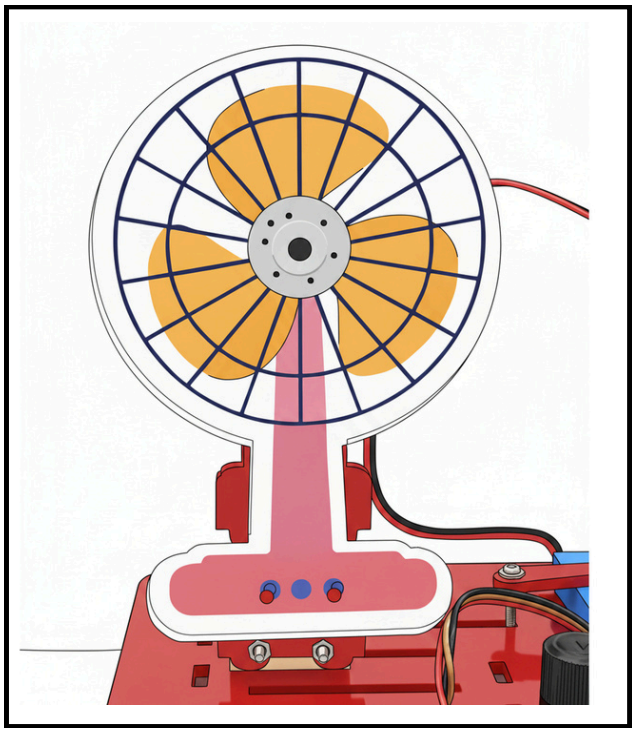
**Step 3-** Fix the dc motor stand to the board and attach the dc motor to it

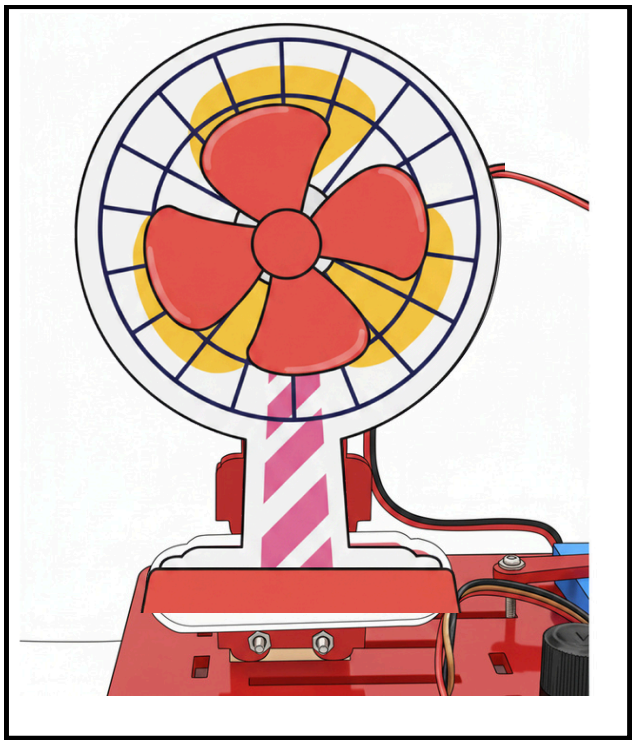


**Step 4-** Attach the fan to the motor as shown



**Step 5-** Attach the template to the stand as shown





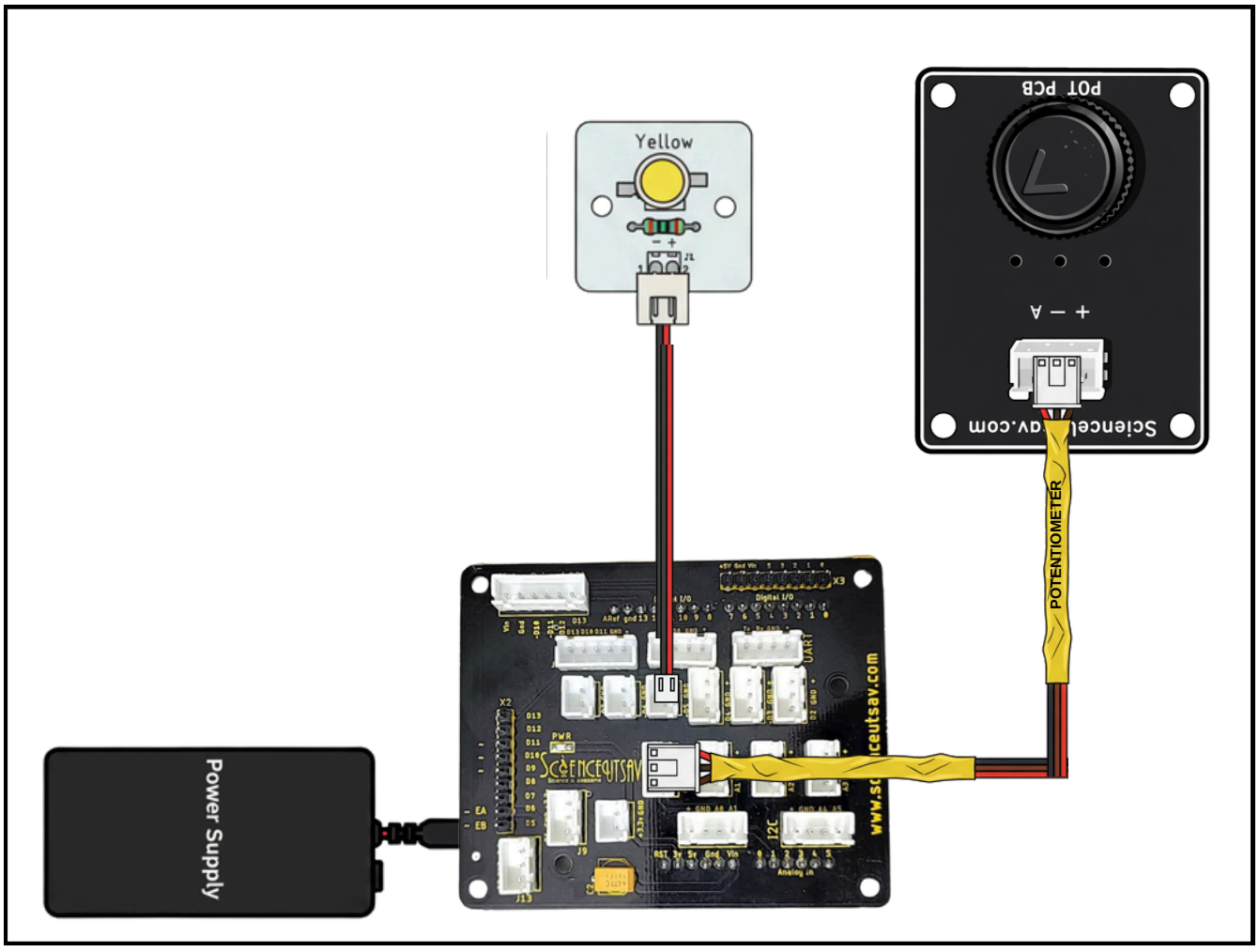
**Your project is now ready for exploration!**

# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and custom switch PCB logic.
*/

const int potSensor = A0;
const int dcMotor = 5;

void setup() {
  pinMode(dcMotor, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int potValue = analogRead(potSensor);
  int motorSpeed = map(potValue, 0, 1023, 0, 255);

  analogWrite(dcMotor, motorSpeed);

  Serial.println(potValue);
  delay(50);
}
```

# BEGINNER PROJECT

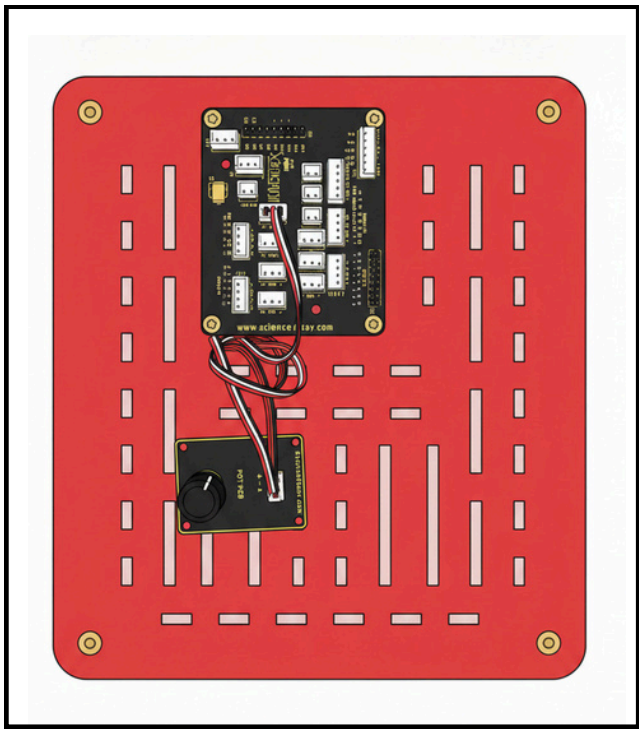
The circuit diagram of the project is given below



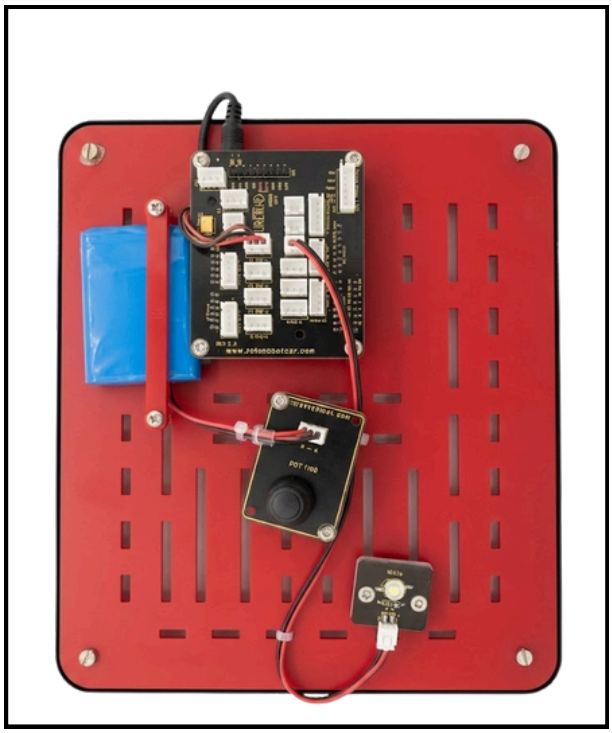

## CONNECTION:

1. 1W LED IN D5
2. POTENTIOMETER PCB IN A0

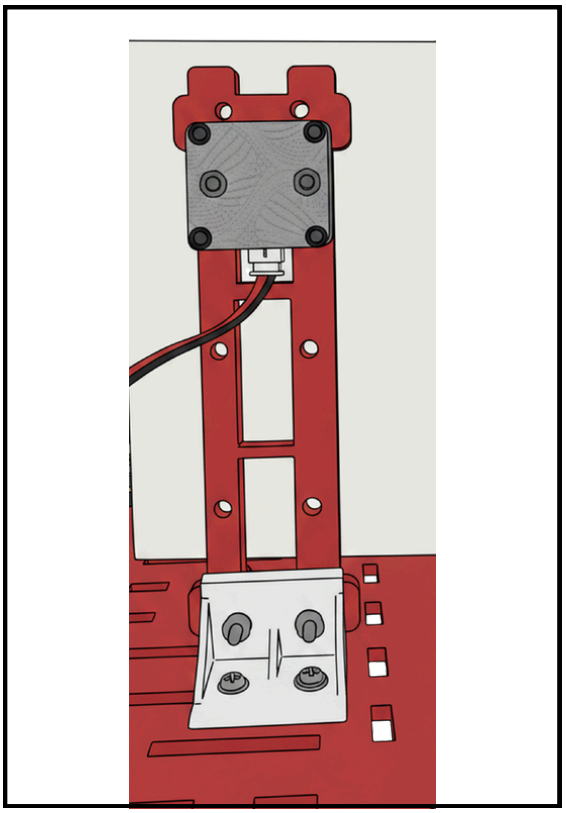
**Step 1-** Fix the potentiometer PCB to the board



**Step 2-** Connect the LED as shown

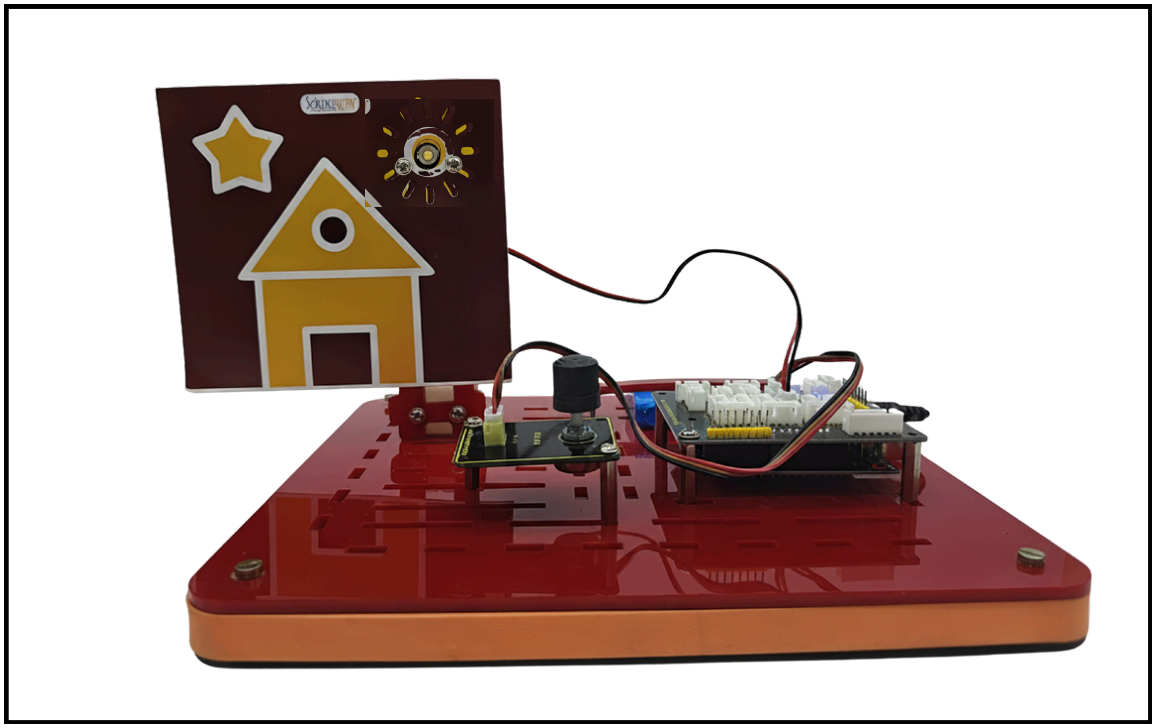


**Step 3-** Attach the stand and connect the LED to it. Secure it with bolt



**Step 4-** Attach the template using the bolts as show

**Step 5-** Attach the LED along with the template



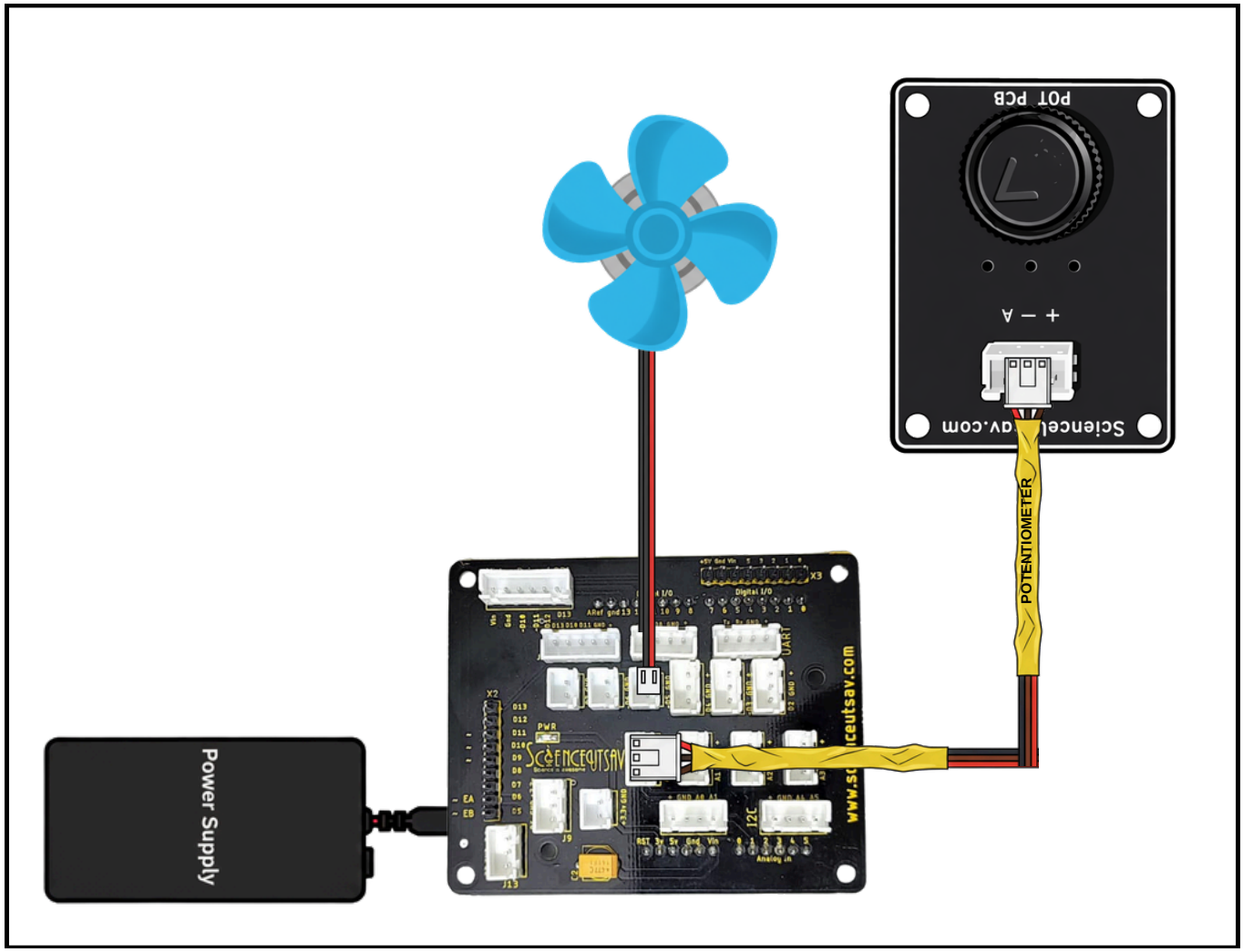
**Now watch the sun light up hour house!**

# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and custom switch PCB logic.
*/

const int potSensor = A0;
const int oneWLed = 5;

void setup() {
  pinMode(oneWLed, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int potValue = analogRead(potSensor);
  int brightness = map(potValue, 0, 1023, 0, 255);

  analogWrite(oneWLed, brightness);

  Serial.println(potValue);
  delay(50);
}
```

# INTERMEDIATE PROJECT

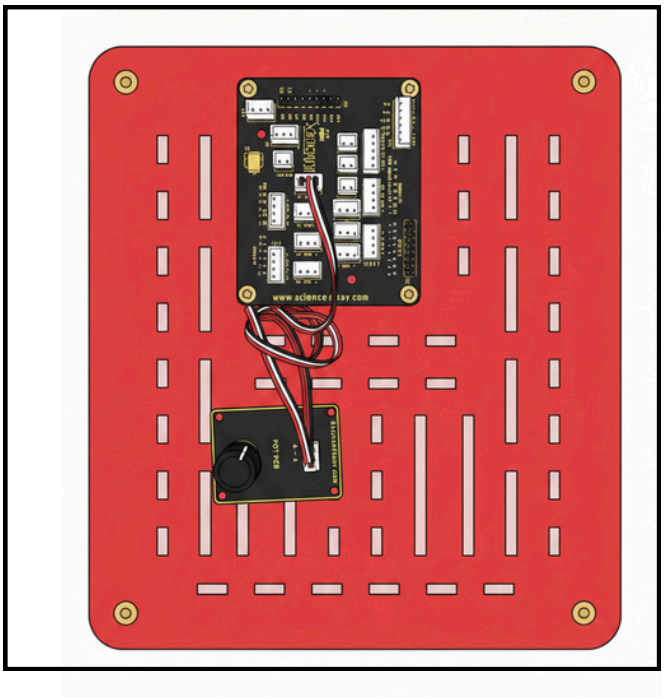
The circuit diagram of the project is given below



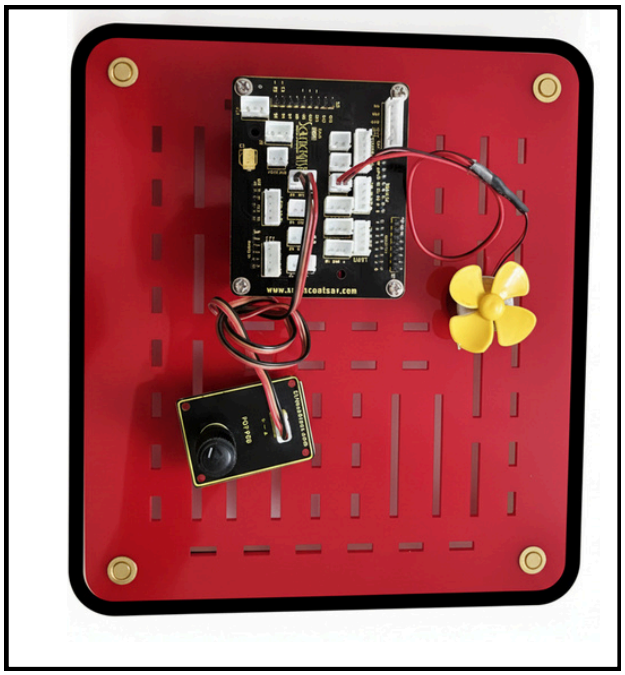

## CONNECTION:

1. DC MOTOR IN D5
2. POTENTIOMETER PCB IN A0

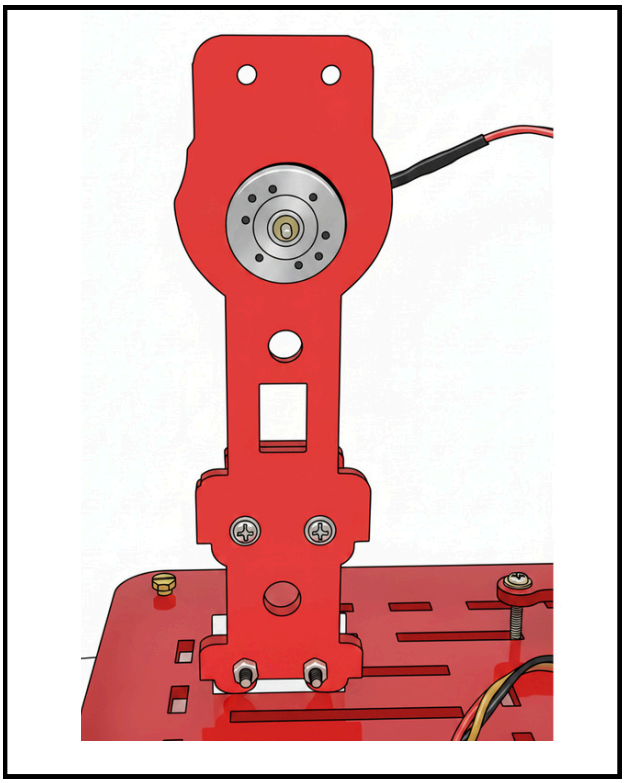
**Step 1-** Fix the potentiometer PCB to the board



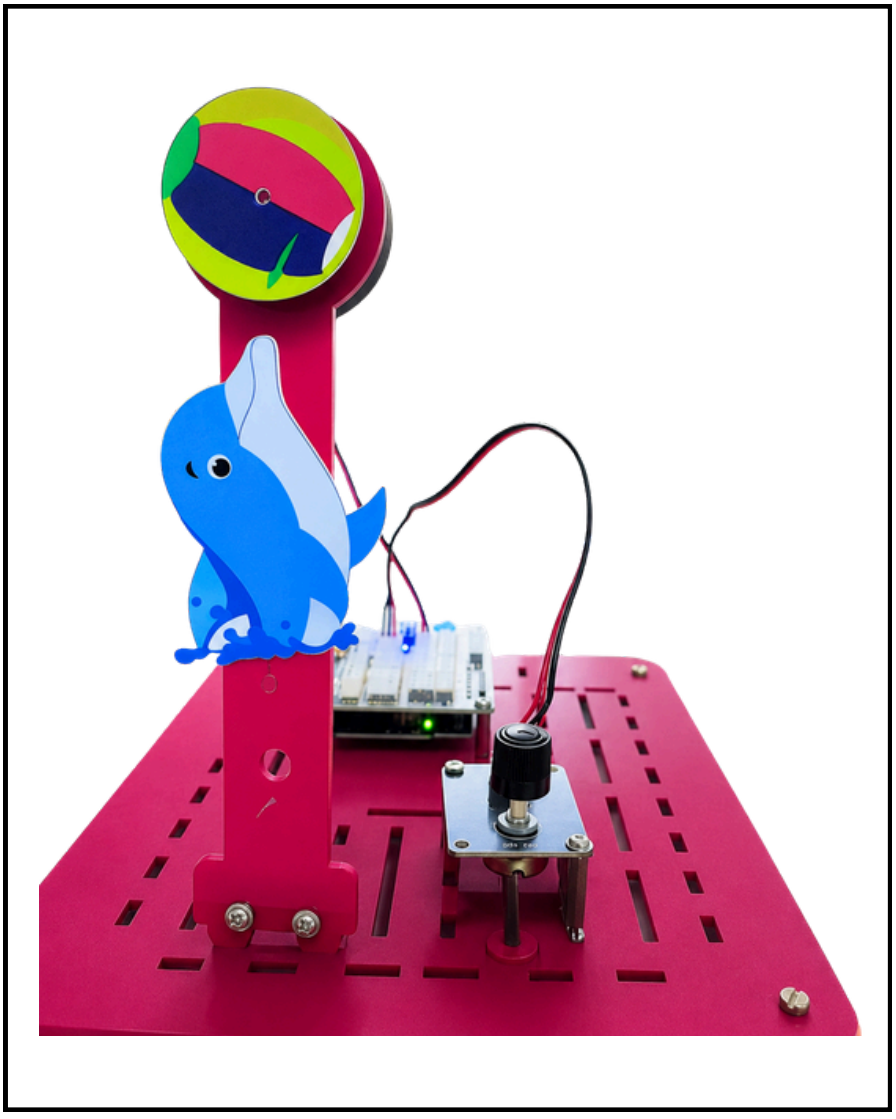
**Step 2-** Connect the dc motor as shown



**Step 3-** Fix the dc motor stand to the board and attach the dc motor to it



**Step 4-** Fix the template to the stand as shown



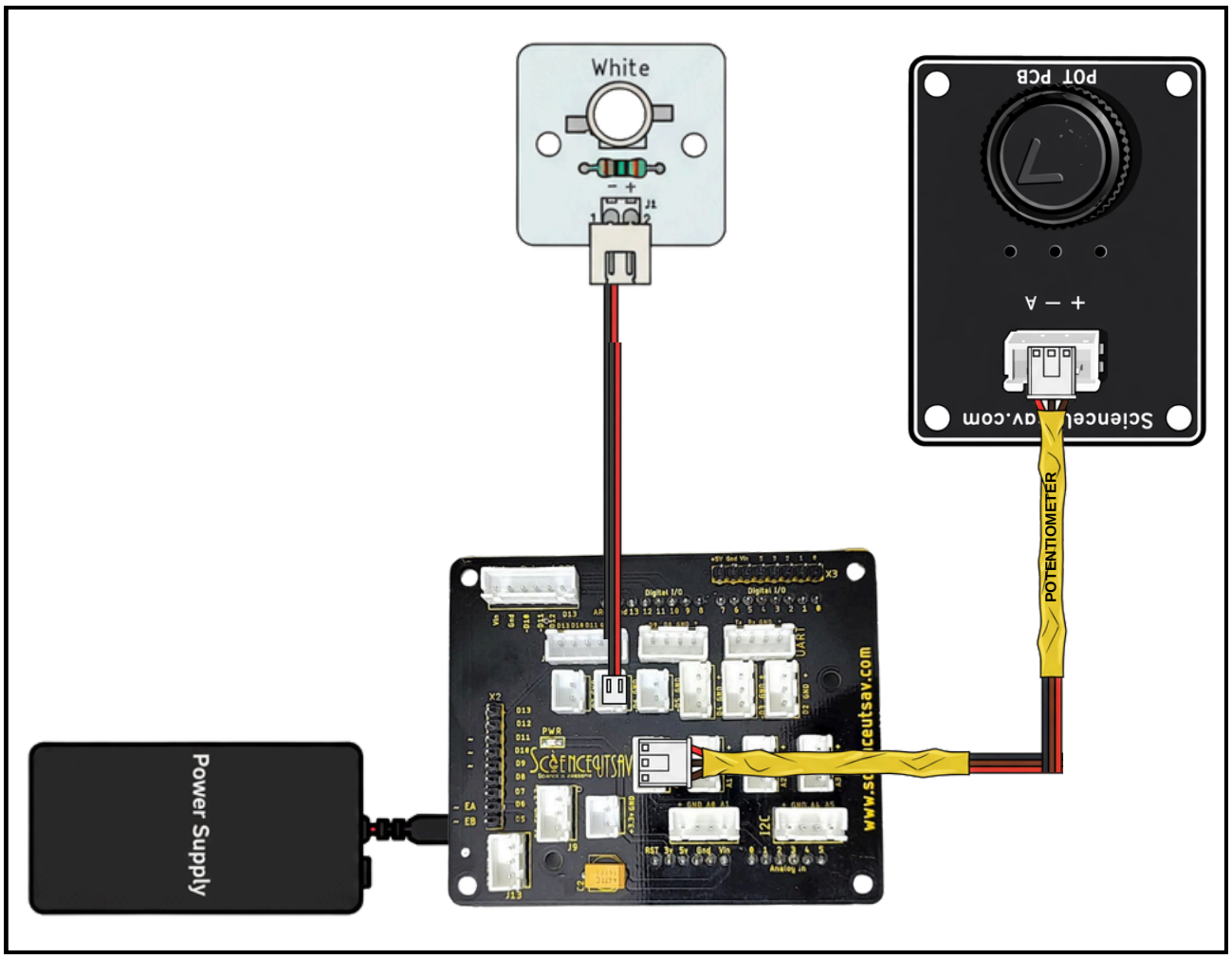
**Now watch your dolphin play with the ball!**

# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and custom switch PCB logic.
*/

const int potSensor = A0;
const int dcMotor = 5;

void setup() {
  pinMode(dcMotor, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int potValue = analogRead(potSensor);
  int motorSpeed = map(potValue, 0, 1023, 0, 255);

  analogWrite(dcMotor, motorSpeed);

  Serial.println(potValue);
  delay(50);
}
```

# INTERMEDIATE PROJECT

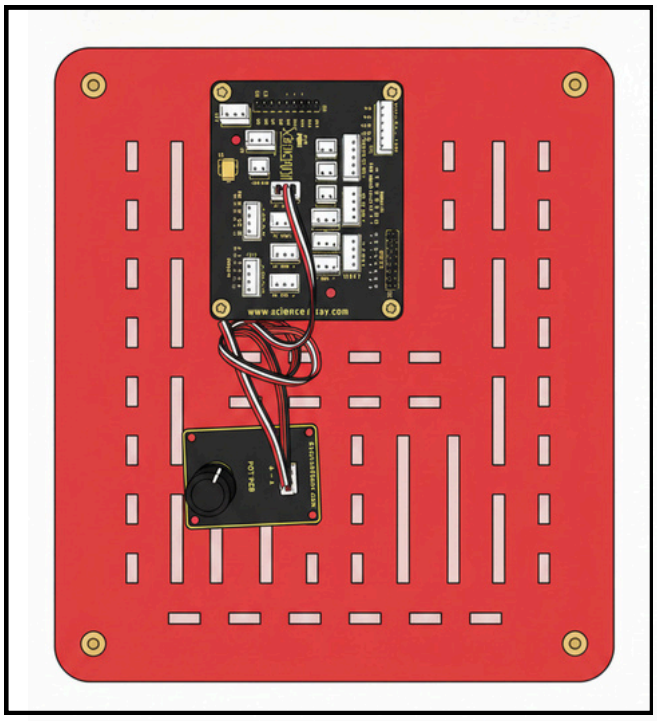
The circuit diagram of the project is given below



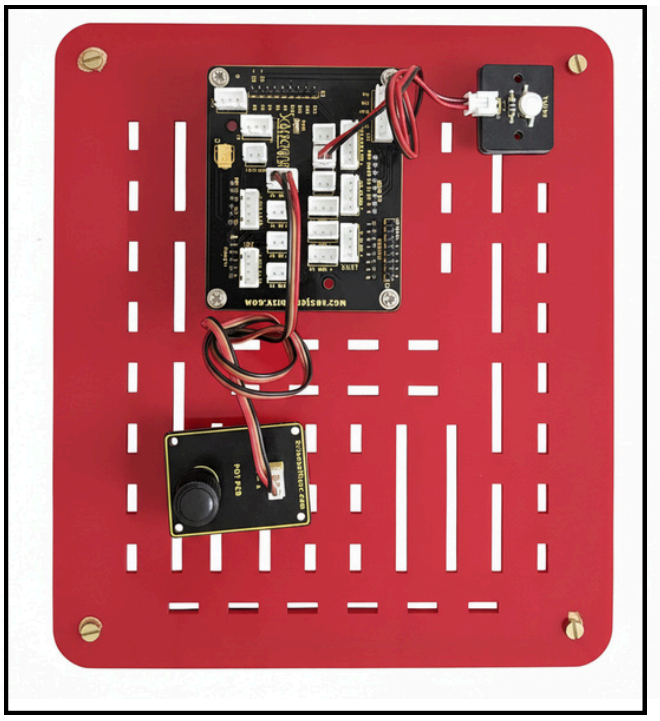

## CONNECTION:

1. 1W LED IN D5
2. POTENTIOMETER PCB IN A0

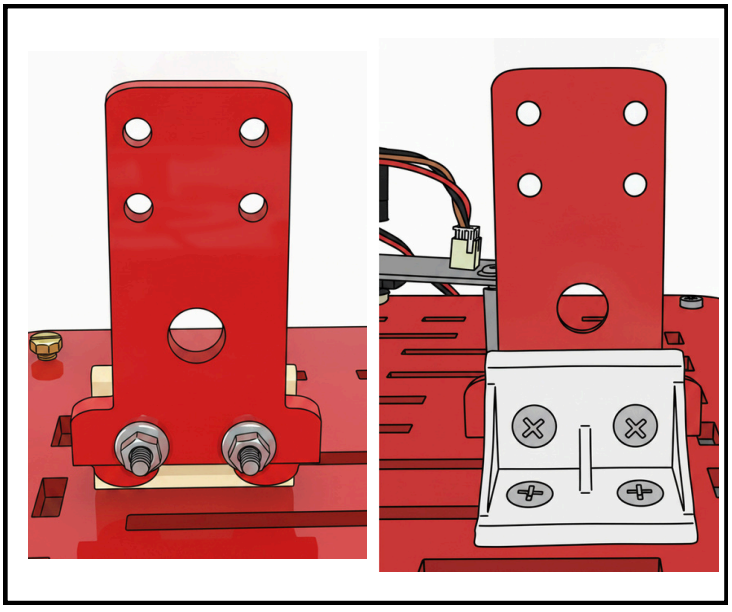
**Step 1-** Fix the potentiometer PCB to the board



**Step 2-** Connect the LED as shown

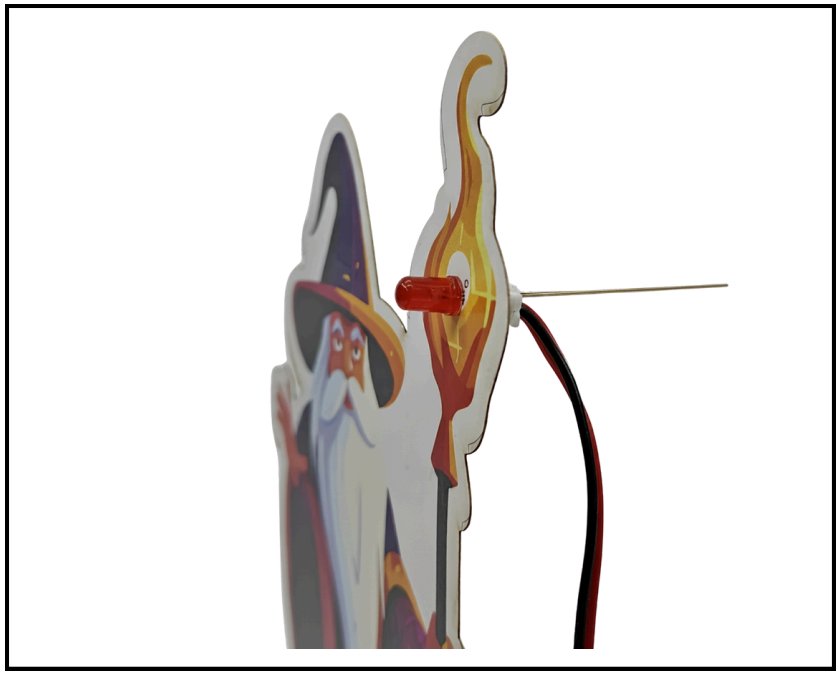


**Step 3-** Attach the stand and secure it with bolt

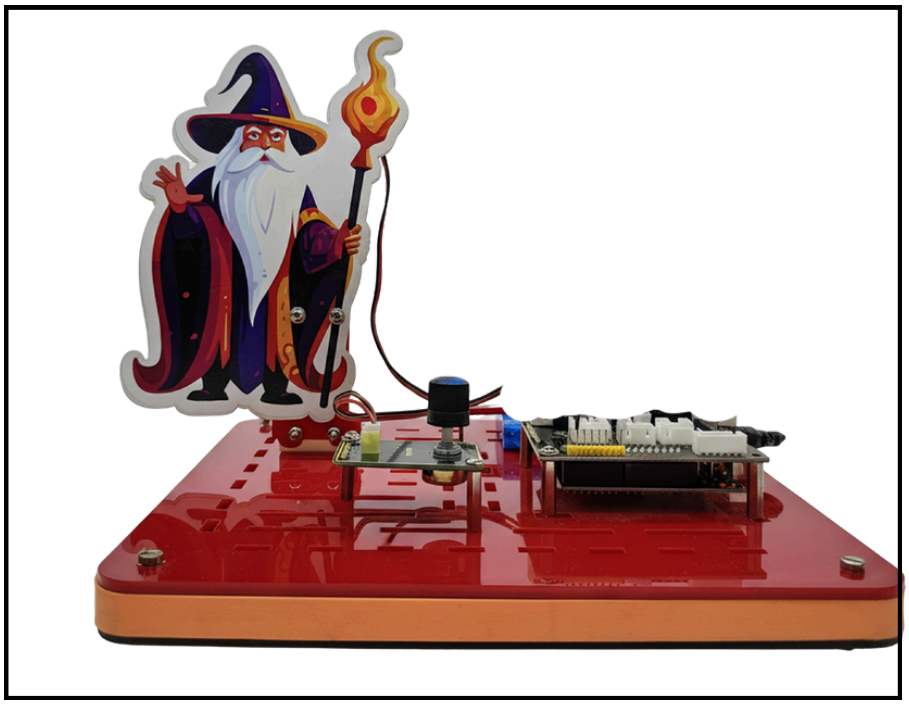


**Step 4-** Attach the template using the bolts as show

**Step 5-** Attach thedip LED to the template and then insert the wire from the back



**Remember:** the long led goes to the red wire andthe short leg goes to the black wire



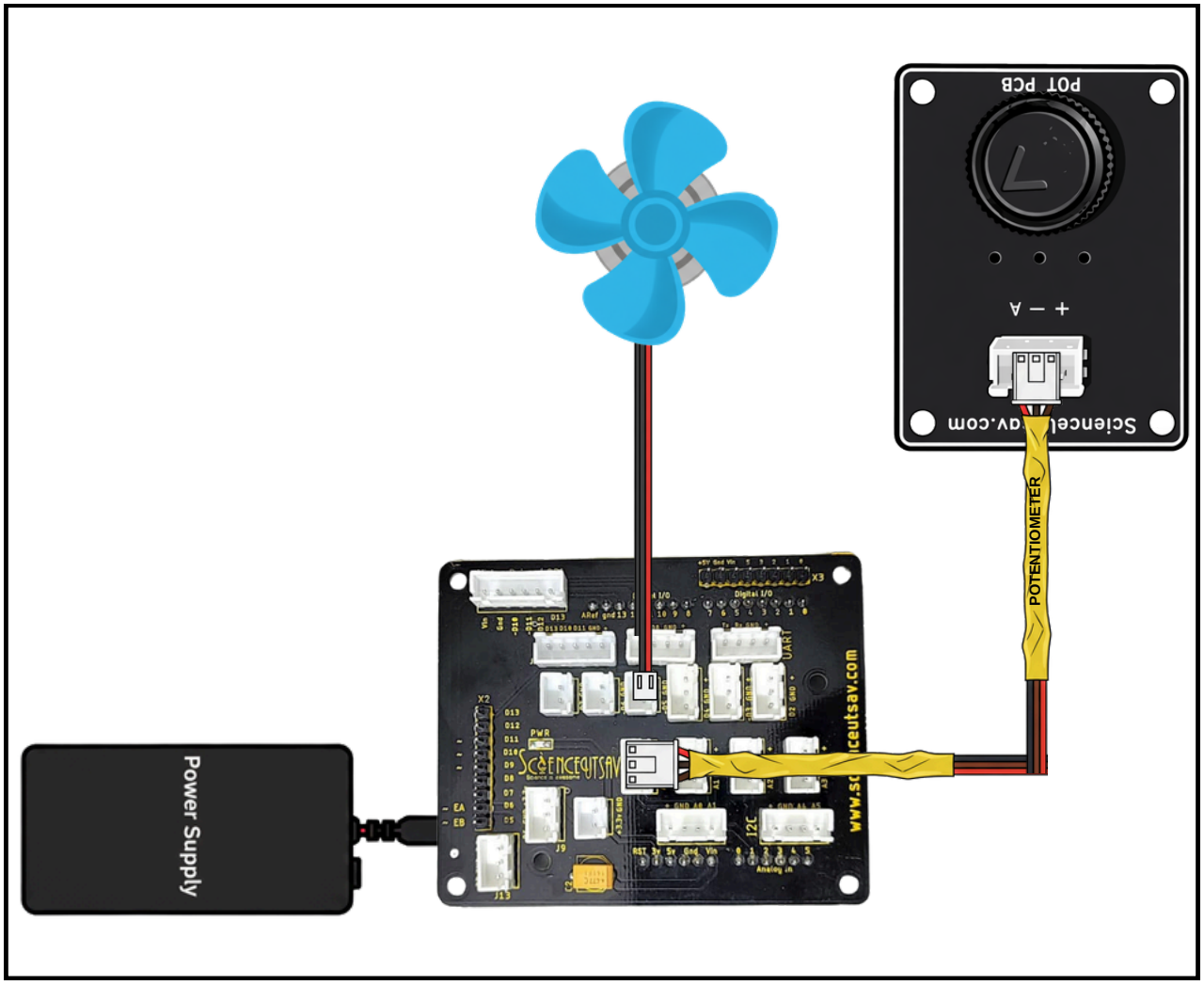
**Enjoy the wizard cast curses with his staff!**

# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and custom switch PCB logic.
*/

const int potSensor = A0;
const int oneWLed = 6;

void setup() {
  pinMode(oneWLed, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int potValue = analogRead(potSensor);
  int brightness = map(potValue, 0, 1023, 0, 255);

  analogWrite(oneWLed, brightness);

  Serial.println(potValue);
  delay(50);
}
```

# ADVANCE PROJECT

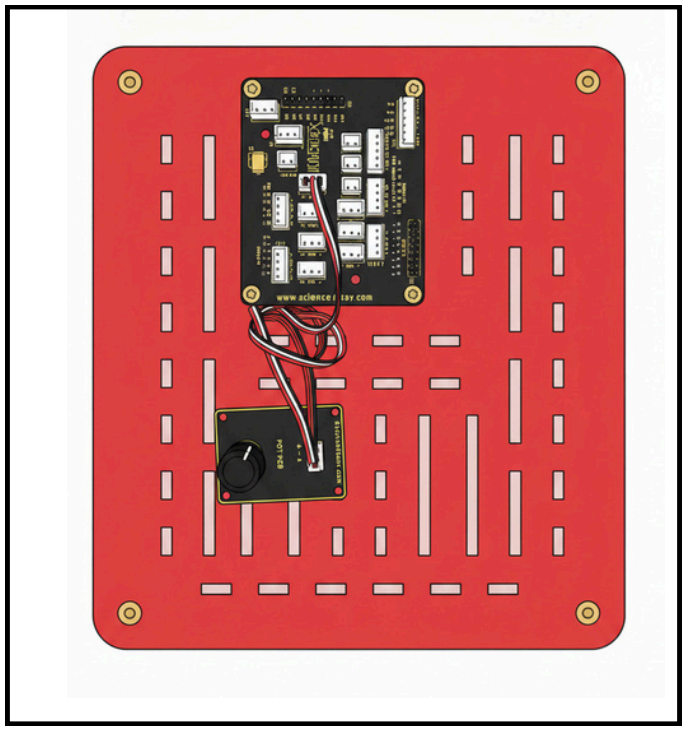
The circuit diagram of the project is given below



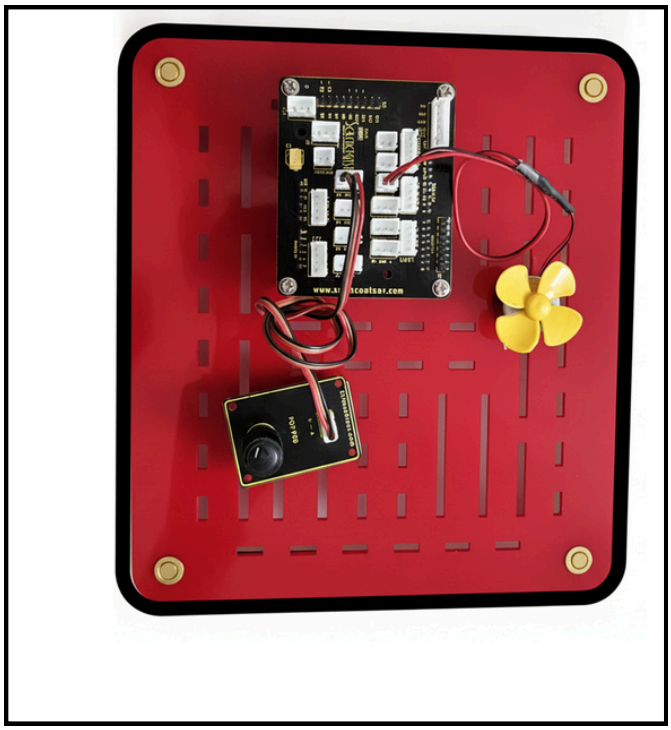

## CONNECTION:

1. DC MOTOR IN D5
2. POTENTIOMETER PCB IN A0

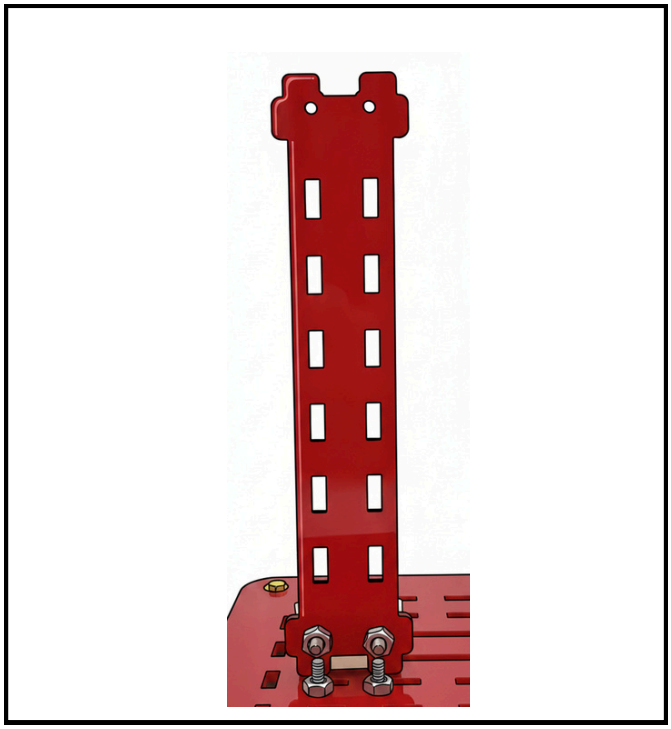
**Step 1-** Fix the potentiometer PCB to the board



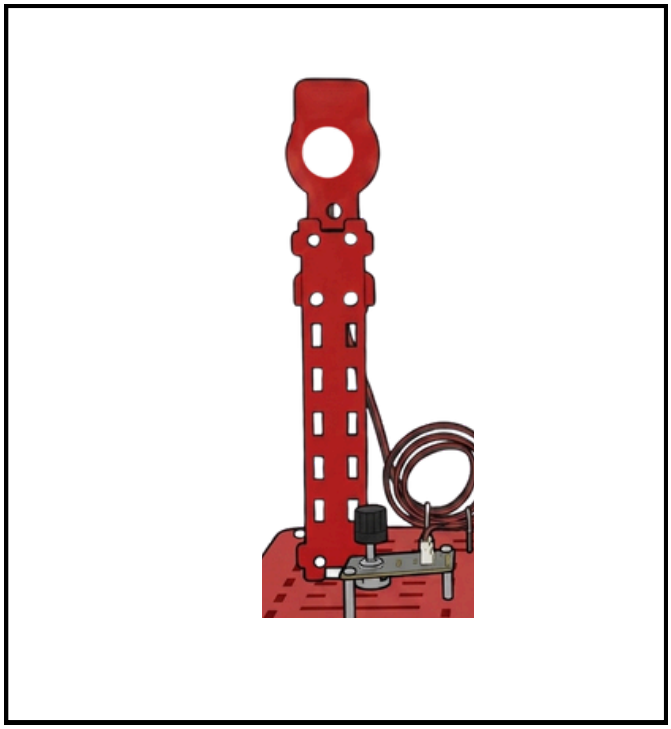
**Step 2-** Connect the dc motor as shown



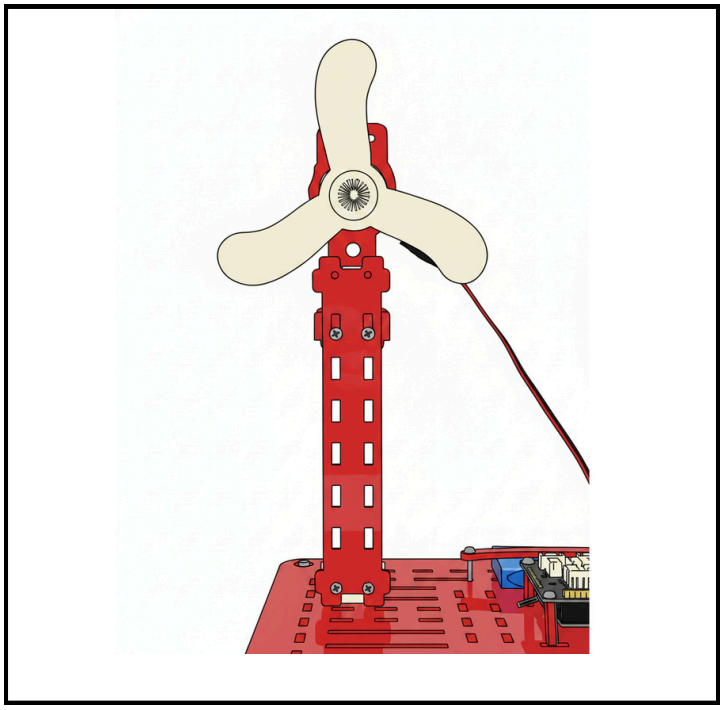
**Step 3-** Fix the first stand onto the board

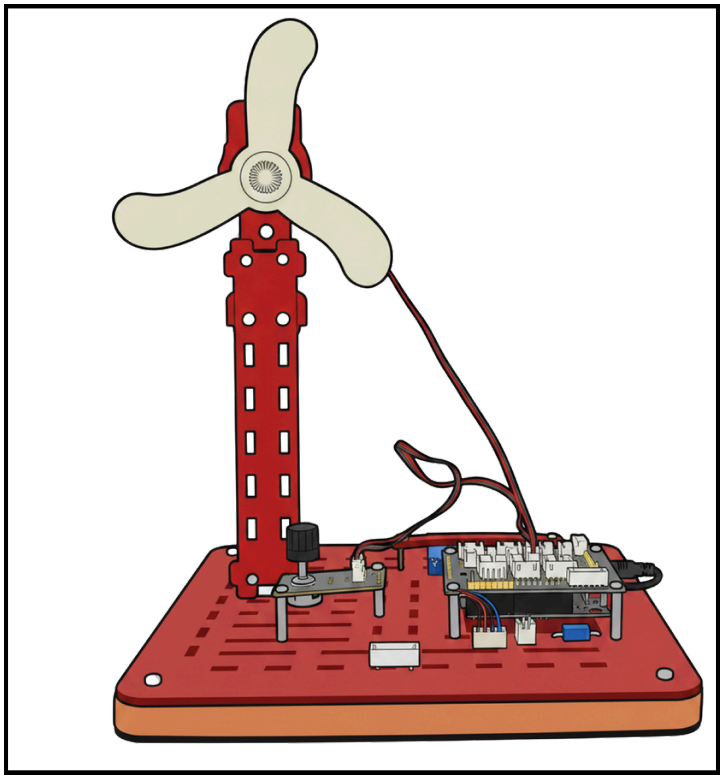


**Step 4-** Fix the second stand on top of the first as shown



**Step 5-** Fix the dc motor and the fan onto the stand





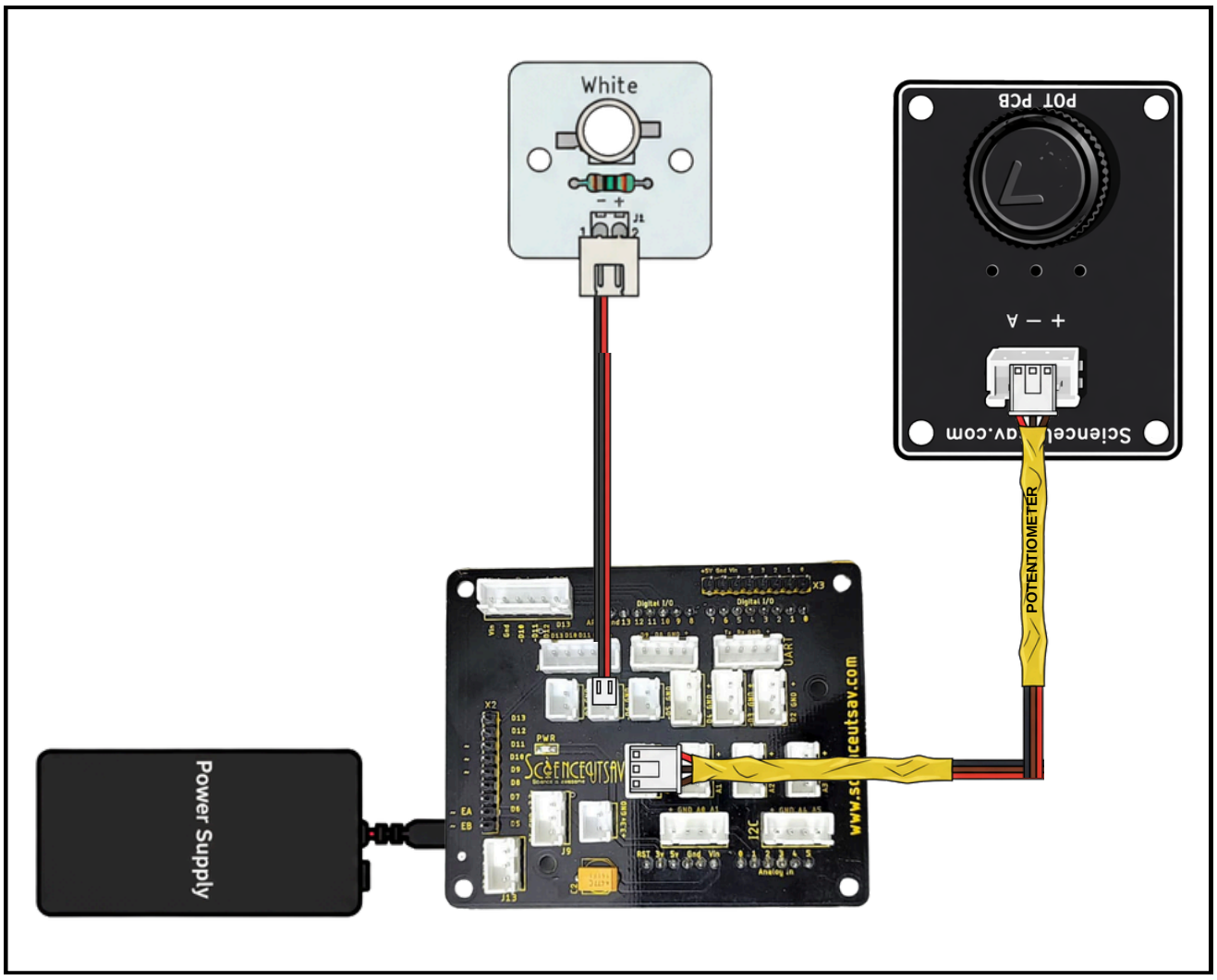
**Your windmill is now ready to turn ideas into energy!**

# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and custom switch PCB logic.
*/

const int potSensor = A0;
const int dcMotor = 5;

void setup() {
  pinMode(dcMotor, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int potValue = analogRead(potSensor);
  int motorSpeed = map(potValue, 0, 1023, 0, 255);

  analogWrite(dcMotor, motorSpeed);

  Serial.println(potValue);
  delay(50);
}
```

# ADVANCE PROJECT

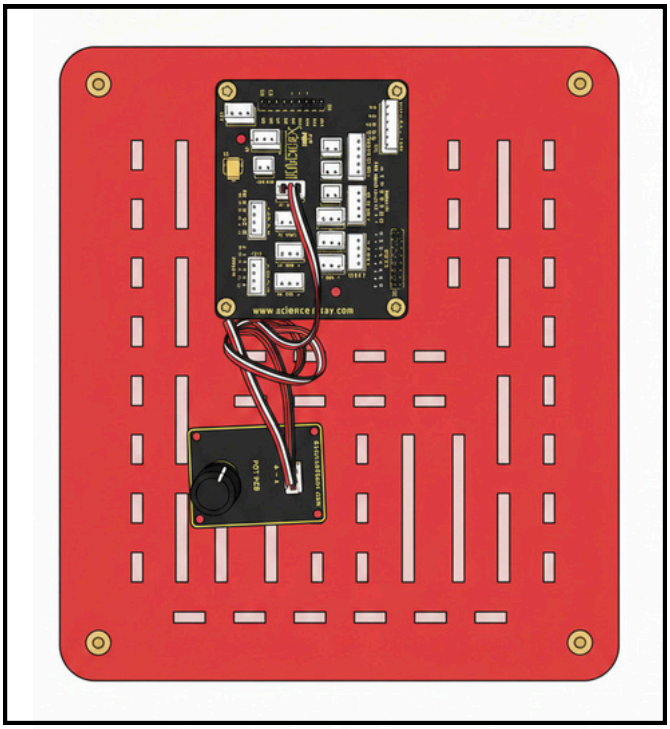
The circuit diagram of the project is given below



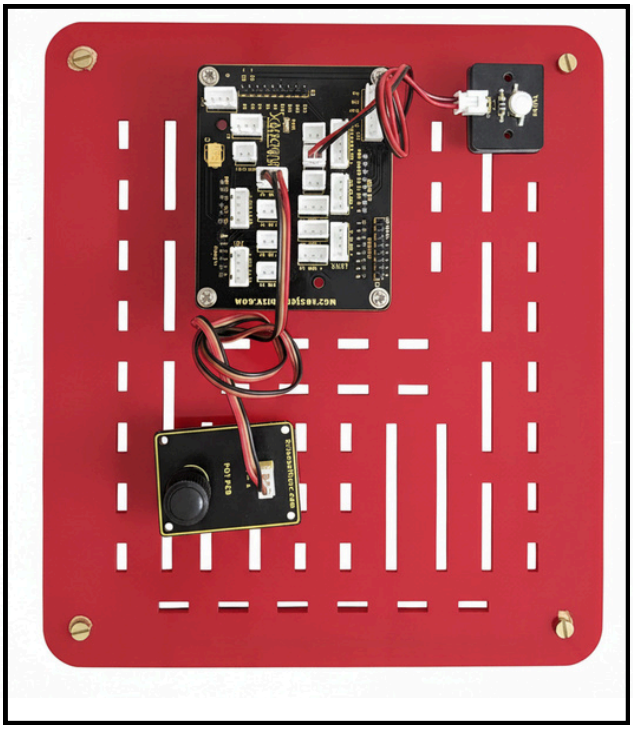

## CONNECTION:

1. 1W LED IN D6
2. POTENTIOMETER PCB IN A0

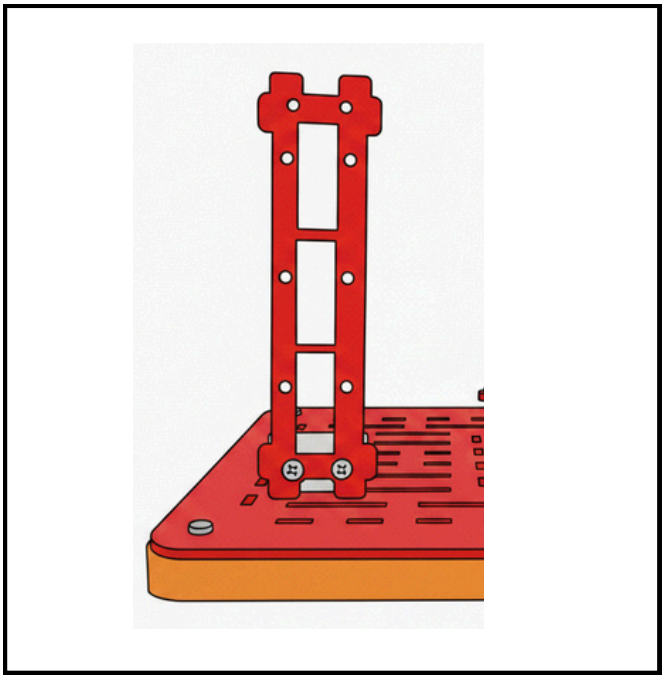
**Step 1-** Fix the potentiometer PCB to the board



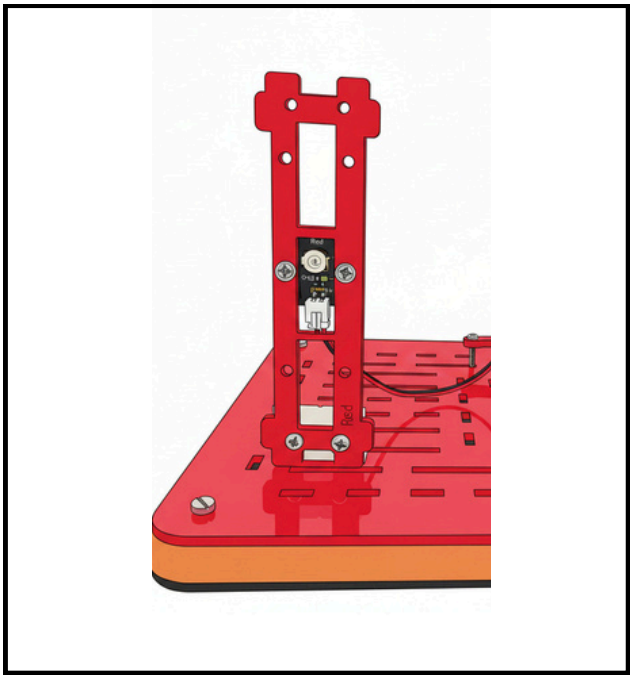
**Step 2-** Connect the LED as shown



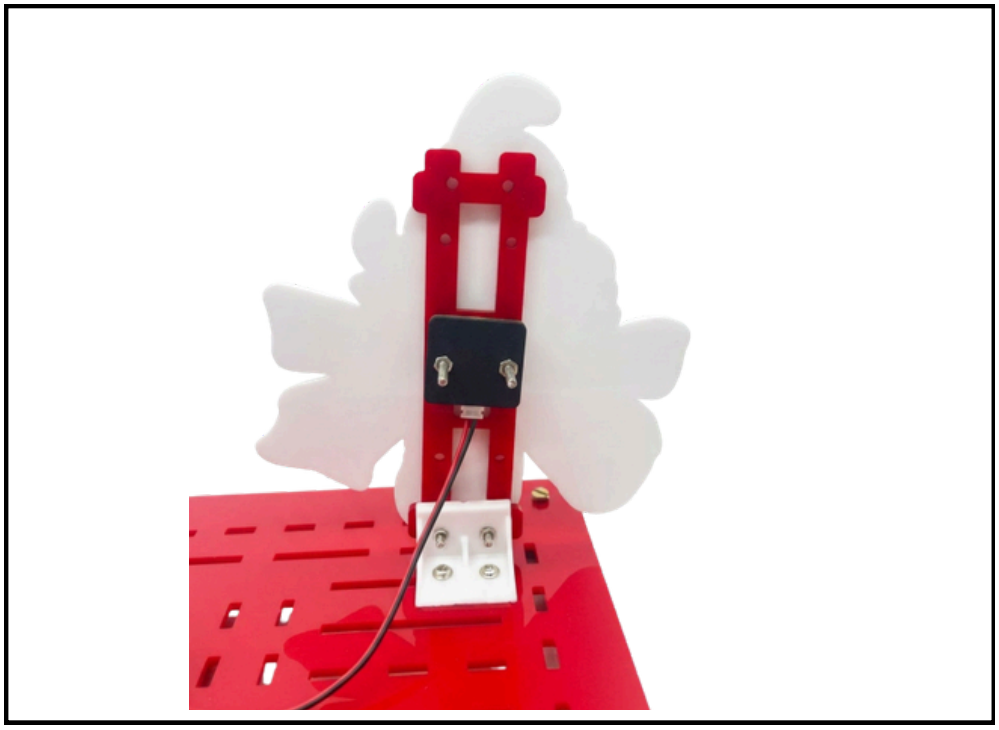
**Step 3-** Fix the stand onto the board

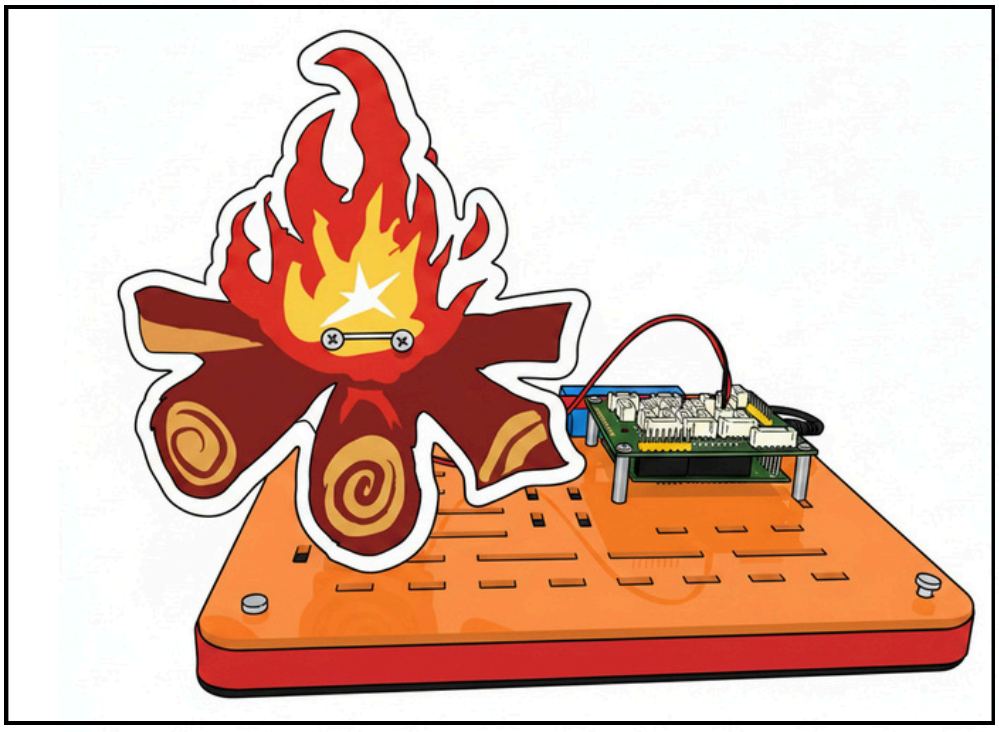


**Step 4-** Attach the led to the stand using bolts



**Step 5-** Attach the template to the stand





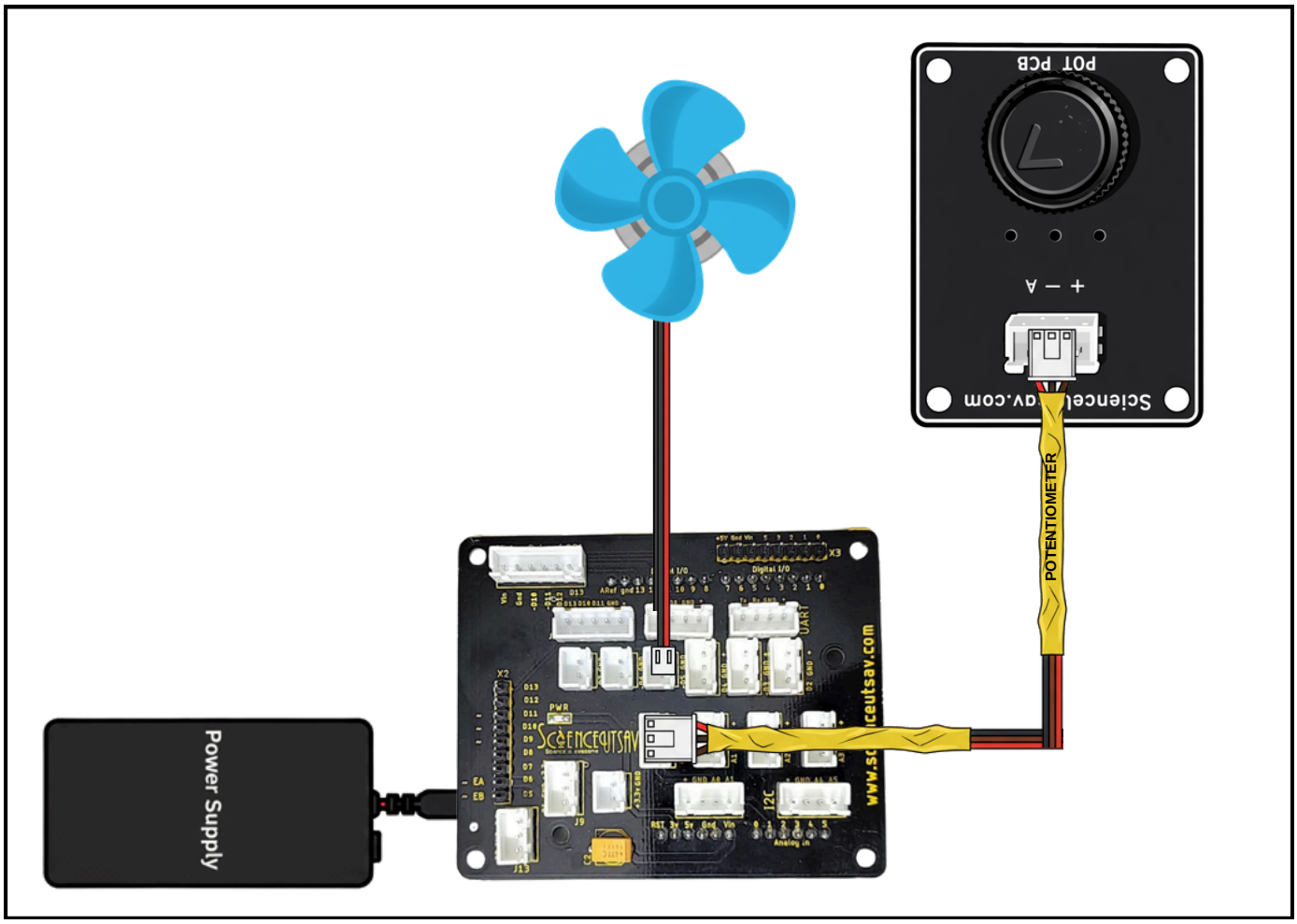
**Enjoy the glow of your campfire!**

# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and custom
switch PCB logic.
*/

const int potSensor = A0;
const int oneWLed = 6;

void setup() {
  pinMode(oneWLed, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int potValue = analogRead(potSensor);
  int brightness = map(potValue, 0, 1023, 0, 255);

  analogWrite(oneWLed, brightness);

  Serial.println(potValue);
  delay(50);
}
```

# EXPERT PROJECT

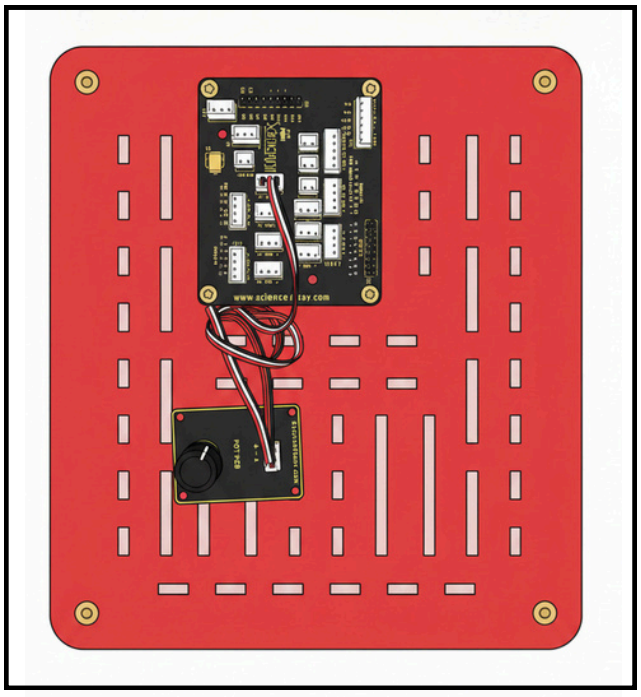
The circuit diagram of the project is given below



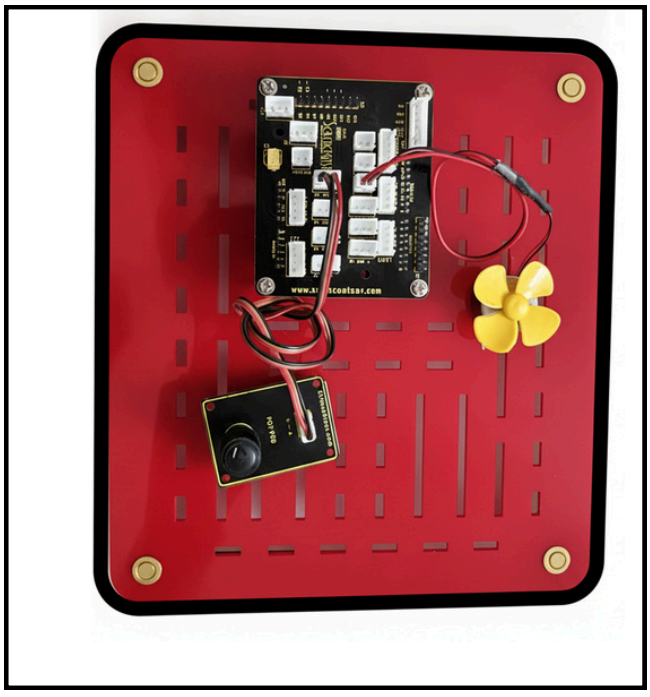

## CONNECTION:

1. DC MOTOR IN D5
2. POTENTIOMETER PCB IN A0

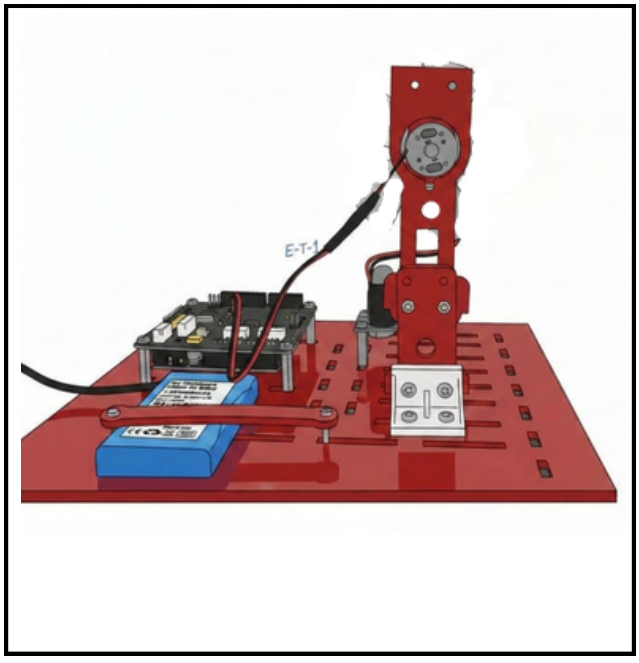
**Step 1-** Fix the potentiometer PCB to the board



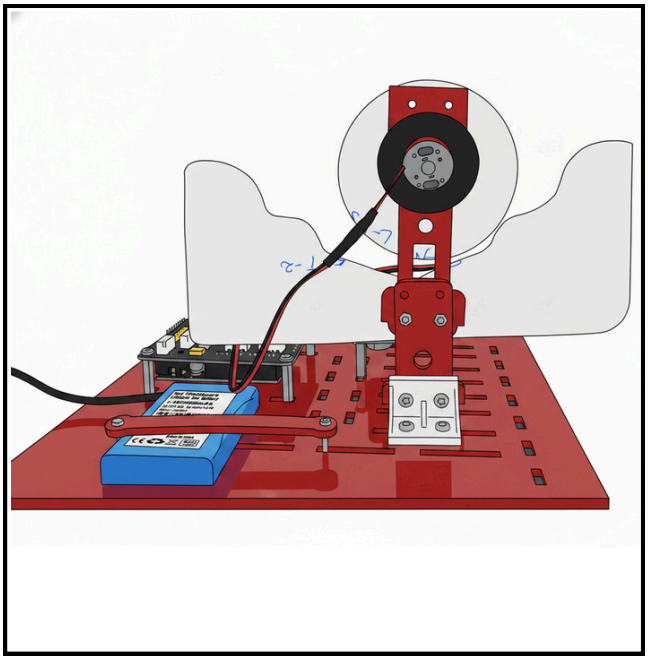
**Step 2-** Connect the dc motor as shown



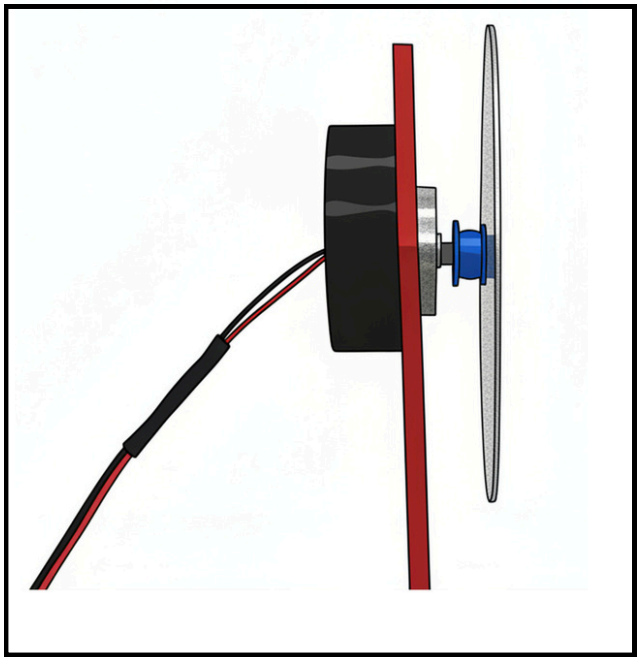
**Step 3-** Fix the stand as shown and connect the dc motor

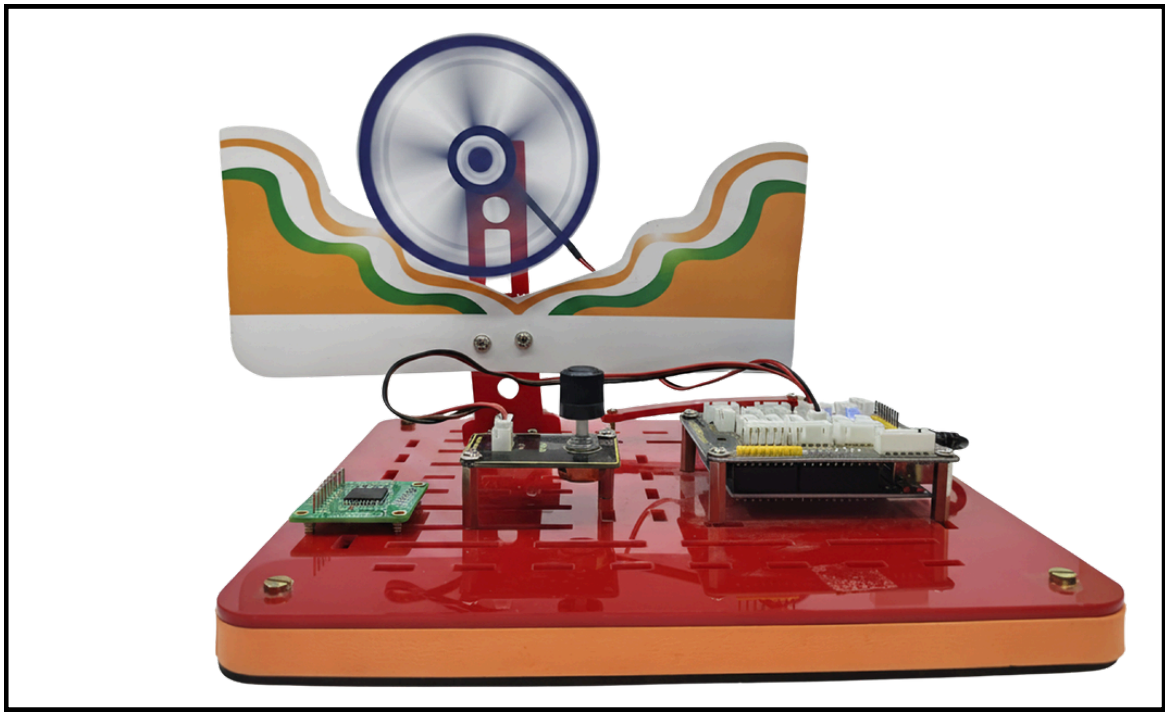


**Step 4-** Attach the template as shown



## **Step 5-** Attach the fan to the motor





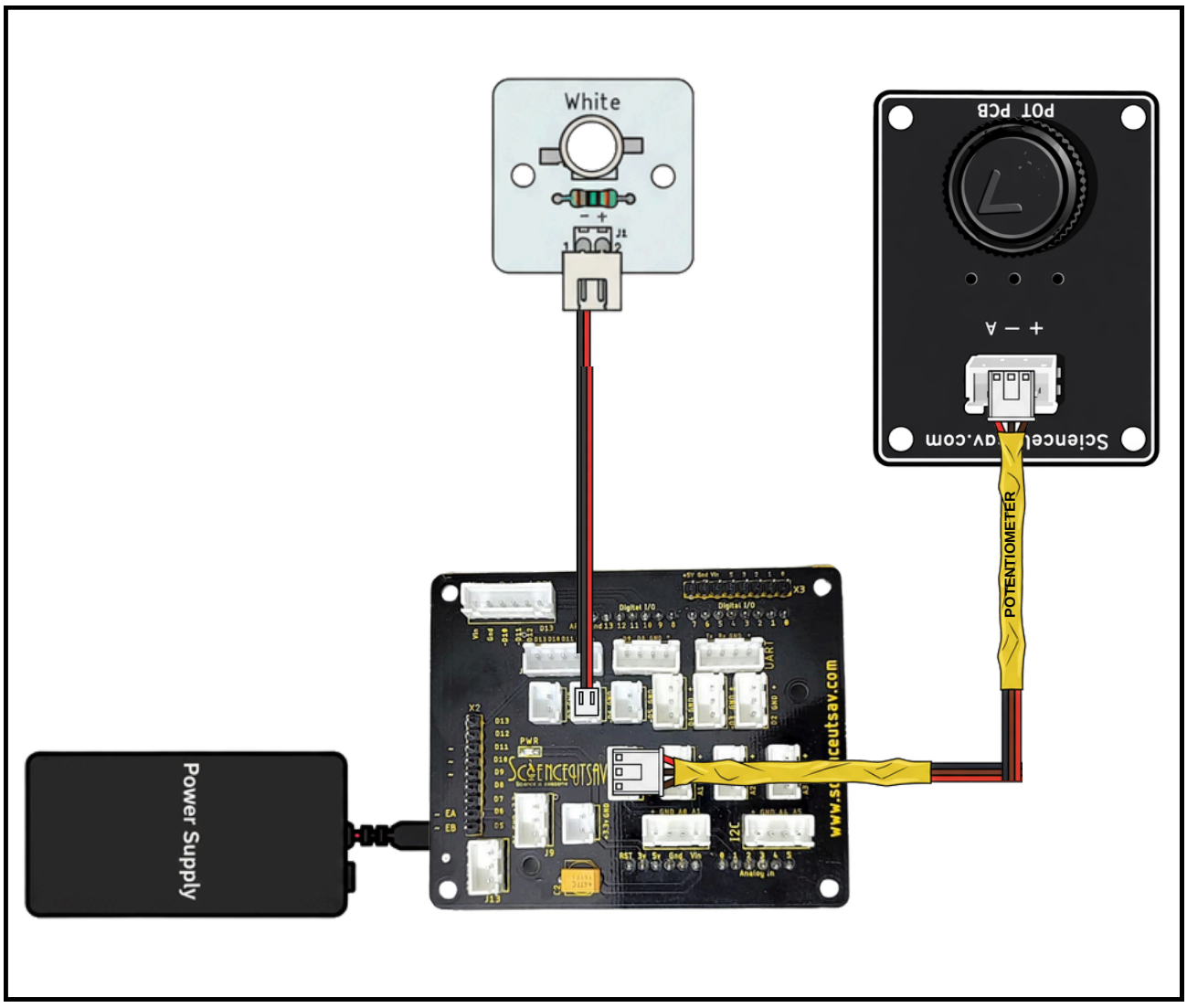
**Your triclour is now ready to spin with pride!**

# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and custom switch PCB logic.
*/

const int potSensor = A0;
const int dcMotor = 5;

void setup() {
  pinMode(dcMotor, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int potValue = analogRead(potSensor);
  int motorSpeed = map(potValue, 0, 1023, 0, 255);

  analogWrite(dcMotor, motorSpeed);

  Serial.println(potValue);
  delay(50);
}
```

# EXPERT PROJECT

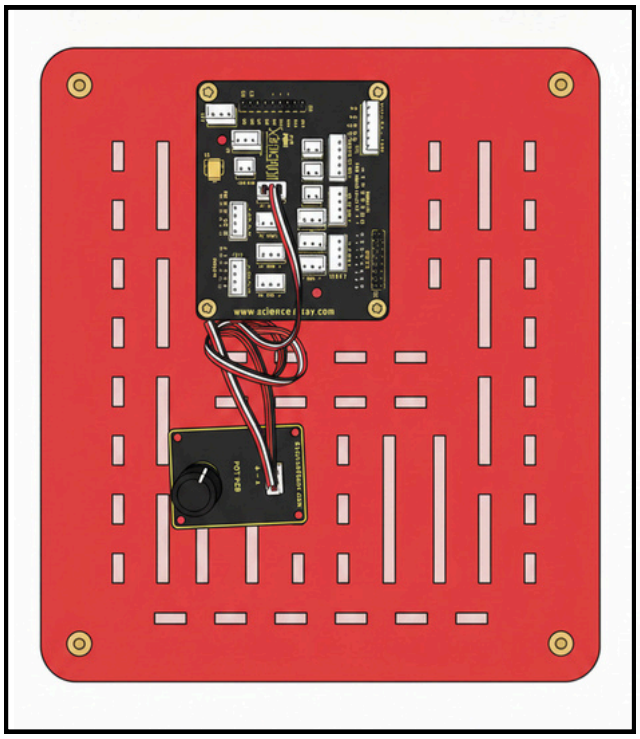
The circuit diagram of the project is given below



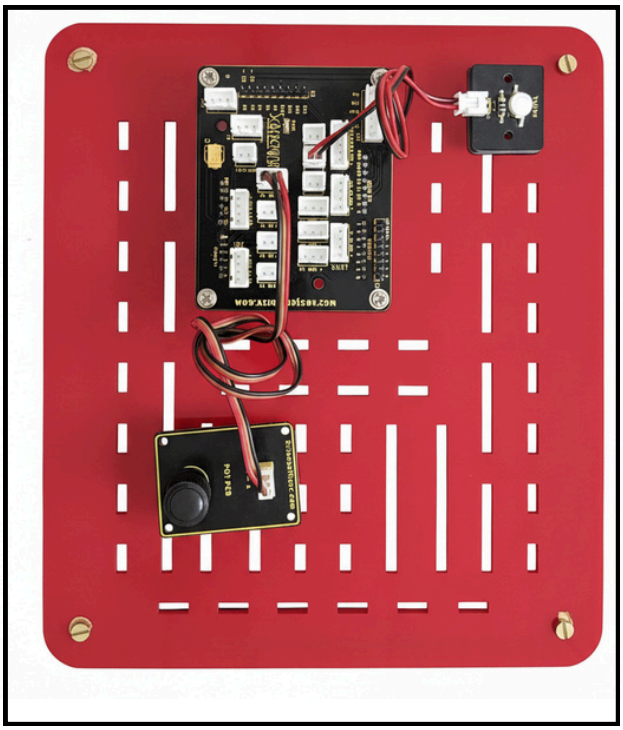

## CONNECTION:

1. 1W LED IN D6
2. POTENTIOMETER PCB IN A0

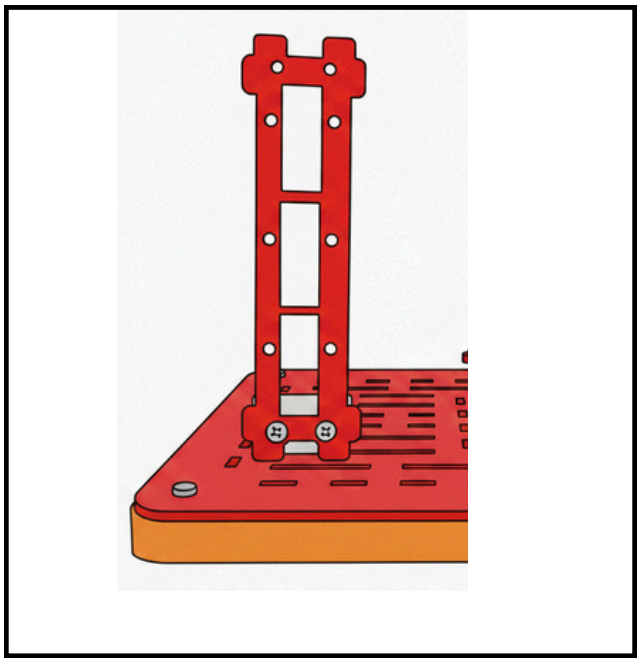
**Step 1-** Fix the potentiometer PCB to the board



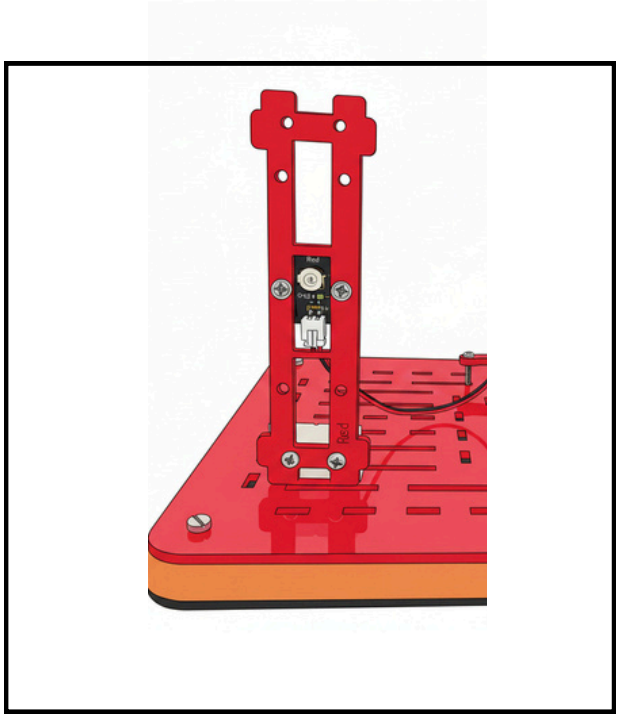
**Step 2-** Connect the LED as shown



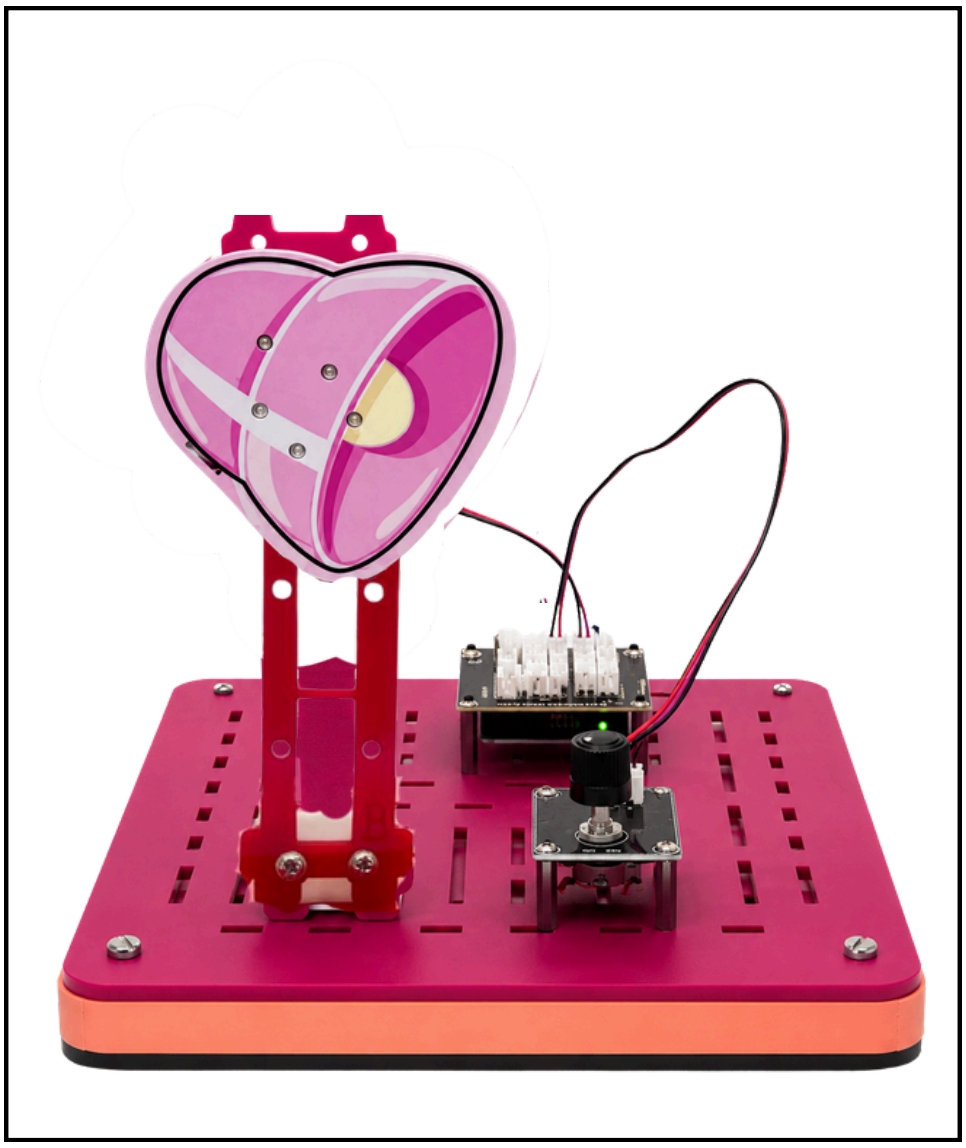
**Step 3-** Fix the stand onto the board



**Step 4-** Attach the led to the stand using bolts



**Step 5-** Attach the template to the stand as shown



**Your lamp is now ready to glow!**

# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and custom
switch PCB logic.
*/

const int potSensor = A0;
const int oneWLed = 6;

void setup() {
  pinMode(oneWLed, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int potValue = analogRead(potSensor);
  int brightness = map(potValue, 0, 1023, 0, 255);

  analogWrite(oneWLed, brightness);

  Serial.println(potValue);
  delay(50);
}
```